Chapter 4

Mixed-Integer Program Modeling

Modeling in optimization is the process of capturing a meaningful approximation of a real life situation into a mathematical representation. Modeling is both "easy" and "difficult." The easy part is formulating mathematical expressions from verbal or written conditions and goals. The hard part is meaningfully capturing real life. We will focus mostly on the easier part. The harder part becomes easier with experience and hard work.

This dichotomy between the easy and difficult is illustrated with a bit of humor by George Dantzig (who developed the Simplex Method to solve linear programs) in his article "The Diet Problem," *Interfaces*, **20:** 4, July-August 1990. Dantzig tells a true story of his own attempts to create a weight-loss diet in the early 1950's for himself. It was a simple matter to create a linear program that maximized "a feeling of fullness" while meeting nutritional guidelines and implementing a calorie restriction. However, the solution(s) to the model were impractical meals which included things like 500 gallons of vinegar, 200 bouillon cubes or 2 pounds of bran. The problem was not Dantzig's ability to model nutritional constraints, nor the Simplex Algorithm's ability to return an optimal solution. The problem was that the constraints (and possibly the objective) did not model Dantzig's diet needs with sufficient accuracy and completeness.

Methodology

The following is a suggested method for modeling optimization problems. This should not be taken as a recipe. Anyone should feel free to model problems in any way that makes sense and results in a well-formulated and notationally correct optimization problem.

1. **Decision Variables.** Identify those problem quantities over which the decision-maker has some control, or ability to choose. All other problem quantities will be written in terms of these decision variables. You can always add more decisions

variables as you continue to model a problem, but identifying key variables at the beginning is important.

2. **Objective Function.** Construct your goal statement in terms of your decision variables. What quantity are you trying to maximize or minimize? Sometimes the objective is not easily written as a linear function, but you should attempt to do so. We will see many examples of nonlinear objective functions which we can linearize when an initial nonlinear model is complete.

3. **Constraints.** List all restrictions imposed on your decision variables. Also express these quantities as linear functions whenever possible. Sometimes in constructing constraints it may be clarifying to add new decision variables. We will see examples of this as well.

4. **Sign and Integer Restrictions.** Don't forget that many decision variables are naturally restricted in sign (non-negativity is the most common). These constraints must be included. Remember that any solution algorithm can only use the information you provide.

5. **MIP in a Standard Form.** Write the optimization problem as a Mixed-Integer Program in an appropriate standard form. This step may be quite involved or relatively simple depending on your model and upon the software you will use to find a solution. Your goal is a model which can be solved by the software of your choice.

A First Modeling Example

Consider the following scenario.

> *Alex and Ryen need to write a collaborative paper for their English class. Alex writes at 3.5 pages per hour and makes 3 grammatical errors per page. Ryen writes at 2.5 pages per hour and makes 1 grammatical error per page. They have discovered that the grade given by the professor is proportional to the number of written pages in the assignment. However, for every 5 grammatical errors the professor deducts an equivalent of 1 written page. The maximum number of grammatical mistakes the professor will tolerate is 50. The total of 8 person-hours is allowed for completing the assignment. Previously scheduled engagements limit Ryen's time participation to 5.5 hours. Formulate a linear program which can be used to find a time budget that will maximize their grade.*

Let's use the suggested modeling methodology to complete the assigned task.

1. **Decision Variables.** Alex and Ryen need to decide, for example, how many hours each needs to spend writing. They cannot change how the professor grades or how many total person-hours are available. So, we can define

$$\begin{aligned} x_1 &= \text{(number of hours that Alex writes)} \\ x_2 &= \text{(number of hours that Ryen writes)} \end{aligned}$$

   Notice that each decision variable is a well-defined numerical quantity with associated units. Some poor definition examples: "x_1 = Alex", "x_1 = Alex writes", "x_1 = time for Alex".

   We could have used a different set of decision variables. For example, Alex and Ryen could be thinking in terms of how many pages each could write and use x_1 and x_2 as the number of pages Alex and Ryen will write, respectively. This choice is equally valid. We will continue with the choice listed above.

2. **Objective Function.** The goal is to maximize the grade of the paper. So, we must construct a function $f : \mathbb{R}^2 \to \mathbb{R}$ for which the inputs are the decision variable values (the number of hours each spends writing) and outputs the "grade value" of the paper.

$$\begin{aligned} z = f(x) &= \text{(grade value assigned to the paper)} \\ &= \text{(effective number of pages written)} \\ &= \text{(number of written pages)} - \text{(page reduction imposed by professor)} \\ &= \text{(pages Alex writes)} - \text{(page reduction due to Alex's errors)} \\ &\quad + \text{(pages Ryen writes)} - \text{(page reduction due to Ryen's errors)} \\ &= \frac{3.5 \text{ pages}}{\text{hour}} x_1 - \frac{1 \text{ page}}{5 \text{ errors}} \cdot \frac{3 \text{ errors}}{\text{page}} \cdot \frac{3.5 \text{ pages}}{\text{hour}} \cdot x_1 \\ &\quad + \frac{2.5 \text{ pages}}{\text{hour}} x_2 - \frac{1 \text{ page}}{5 \text{ errors}} \cdot \frac{1 \text{ errors}}{\text{page}} \cdot \frac{2.5 \text{ pages}}{\text{hour}} \cdot x_2 \\ &= 1.4\ x_1 + 2\ x_2. \end{aligned}$$

   This objective does not actually assign a letter grade, but it does return the effective length of the paper for any choice of hours which Alex and Ryen decide to write.

3. **Constraints.** There are three restrictions on the decision variables that must be met.

   (a) There is a limit on the total number of grammatical errors which the

professor will tolerate.

$$\begin{aligned}
50 &\geq \text{(total number of errors)} \\
&= \text{(number of errors in Alex's writing)} \\
&\quad + \text{(number of errors in Ryen's writing)} \\
&= \frac{3 \text{ errors}}{\text{page}} \cdot \frac{3.5 \text{ pages}}{\text{hour}} \cdot x_1 + \frac{1 \text{ errors}}{\text{page}} \cdot \frac{2.5 \text{ pages}}{\text{hour}} \cdot x_2 \\
&= 10.5\ x_1 + 2.5\ x_2.
\end{aligned}$$

or equivalently,

$$10.5\ x_1 + 2.5\ x_2 \leq 50.$$

(b) There is a limit on the total number of person-hours that Alex and Ryen can use to write. That is, if they add the times they each work it cannot exceed the limit of eight hours.

$$\begin{aligned}
8 &\geq \text{(total person-hours spend writing)} \\
&= \text{(hours Alex spends writing)} + \text{(hours Ryen spends writing)} \\
&= x_1 + x_2.
\end{aligned}$$

(c) Ryen has as a maximum number of hours to spend on this writing assignment. This constraint is the most straightforward.

$$x_2 \leq 5.5$$

4. **Sign Restrictions.** Neither Alex nor Ryen can write a negative amount of time:

$$x \geq 0.$$

While these sign constraints may seem unnecessary, such constraints can be critically important to include. It may be possible that if Alex were to write a negative amount of time, this would effectively increase the error budget and effective number of pages. These types of unrealistic situations can occur in optimization problems if the modeler is not careful.

5. **Standard Form.** Finally, we collect all of our model pieces into one place as an optimization problem.

$$\begin{aligned}
\max_{x} \quad & z = f(x) = 1.4\ x_1 + 2x_2 \\
\text{s.t.} \quad & 21x_1 + 5x_2 \leq 100 \\
& x_1 + x_2 \leq 8 \\
& x_2 \leq 5.5 \\
& x \geq 0 \\
& x \in \mathbb{R}^2
\end{aligned}$$

In the above representation, the first constraint was multiplied by 2 simply to display fewer decimals and/or fractions. This model is already a linear program in inequality standard form. We can also write the model in (matrix) inequality standard form in order to expedite our use of software. We have

$$\begin{aligned} \max_{x} \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \\ & x \in \mathbb{R}^2 \end{aligned}$$

$$c = \begin{bmatrix} 1.4 \\ 2 \end{bmatrix}, \; A = \begin{bmatrix} 21 & 5 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \; b = \begin{bmatrix} 100 \\ 8 \\ 5.5 \end{bmatrix}$$

The remaining sections of this chapter present examples of a variety of mixed-integer programming techniques. Each example introduces one or more new concepts and/or strategies. The reader is encouraged to verify the results which are often simply stated in the text. For example, many solutions are provided and the reader is encouraged to use software to verify these results. This chapter also serves as a resource for compiling modeling methods that are useful in a variety of other problems.

As you consider each example, keep in mind the following points.

1. Decision variables, whether initially stated or added during the modeling process, are always defined with numerical values *and* units.

2. Decision variables can be integer valued, binary valued, or continuously valued. The distinction is always explicitly stated.

3. During the constraint modeling or standard form conversion processes, decision variables may be introduced. For example, we have seen that sometimes slack variables are necessary. Many other examples will be shown.

4. A complete MIP, using correct notation (see Definitions 10 and 11), is always presented as the final modeling product.

5. When a solution is presented, both x^* and z^* are provided *and* a brief interpretation of the solution is provided.

4.1 A-1 Furniture

This first modeling example presents a basic profit maximization problem with resource constraints.

The A-1 Furniture factory is tooled to produce four types of furniture pieces: chairs, tables, desks, and dressers. The warehouse holds various quantities of the five necessary raw materials: oak wood, pine wood, glass, wrought iron, and leather. The table below shows the raw material quantities required to produce each type of furniture piece, the raw material quantities in the warehouse, and the expected profit per manufactured item. For example, producing a table requires 8 board feet of oak, 4 kg of iron, one square foot of leather, and results in a profit of \$33. Formulate an integer program whose solution is a production plan for maximizing total expected profit.

		chair	table	desk	dresser	warehouse stock
oak wood	(bdft)	4	8	6	6	2132
pine wood	(bdft)	1	0	3	4	906
glass	(ft^2)	0	0	1	2	330
iron	(kg)	1	4	3	3	945
leather	(ft^2)	2	1	1	0	441
profit	(\$)	21	33	40	50	

Solution. We let $x \in \mathbb{Z}^4$ be the number of of each type of furniture piece to manufacture. In particular,

$$\begin{aligned} x_1 &= \text{number of chairs to manufacture,} \\ x_2 &= \text{number of tables to manufacture,} \\ x_3 &= \text{number of desks to manufacture,} \\ x_4 &= \text{number of dressers to manufacture.} \end{aligned}$$

The expected total profit, z, is the sum of the expected profits for each type of furniture. We know, for example, that each chair provides \$21 in profit, etc.

$$z = 21x_1 + 33x_2 + 40x_3 + 50x_4.$$

There are constraints on the total amount of each resource we have available. For example, the total oak wood used in production cannot exceed the warehouse total.

$$4x_1 + 8x_2 + 6x_3 + 6x_4 \leq 2132.$$

Similarly for the other four resources:

$$
\begin{aligned}
x_1 + 3x_3 + 4x_4 &\leq 906 && \text{(pine)},\\
x_3 + 2x_4 &\leq 330 && \text{(glass)},\\
x_1 + 4x_2 + 3x_3 + 3x_4 &\leq 945 && \text{(iron)},\\
2x_1 + x_2 + x_3 &\leq 441 && \text{(leather)}.
\end{aligned}
$$

We also have the condition that we cannot manufacture negative quantities of any piece of furniture.

$$x_k \geq 0, \quad k = 1, 2, 3, 4.$$

Collecting our results, we have the IP:

$$
\boxed{
\begin{array}{lrrrrrl}
\max & z = & 21x_1 & +33x_2 & +40x_3 & +50x_4 & \\
s.t. & & 4x_1 & +8x_2 & +6x_3 & +6x_4 & \leq 2132\\
 & & x_1 & & +3x_3 & +4x_4 & \leq 906\\
 & & & & x_3 & +2x_4 & \leq 330\\
 & & x_1 & +4x_2 & +3x_3 & +3x_4 & \leq 945\\
 & & 2x_1 & +x_2 & +x_3 & & \leq 441\\
 & & 0 \leq x & & & & \\
 & & x \in \mathbb{Z}^4 & & & &
\end{array}
}
$$

The optimal solution is

$$
\boxed{
\begin{array}{c}
x^* = \begin{bmatrix} 164 & 30 & 82 & 124 \end{bmatrix}^T,\\
z^* = 13{,}914.
\end{array}
}
$$

Thus, the optimal strategy for A-1 Furniture is to manufacture 164 chairs, 30 tables, 82 desks, and 124 dressers. This will lead to a total expected profit of \$13,914. This solution utilizes all available lumber and glass, but leaves 43 kg of iron and one square foot of leather unused. (This last fact is easily determined by computing Ax^* and comparing with b.)

4.2 Greenside Yard & Lawn Care

This modeling example is a cost minimization problem under fixed requirements. Box consraints are used in the formulation.

Greenside Yard & Lawn Care has received a government grant to create temporary jobs through civic improvement projects in their home town. The grant outlines four types of projects: lawn refurbishing, tree removal, community garden establishment, and city park cleanup. Greenside must decide how many of each project to complete. The cost to Greenside for each type of job (in the order listed above) is \$60, \$200, \$750, and \$250. For each job completed $\frac{1}{5}$, $\frac{3}{4}$, 2, and 1 temporary jobs are created, respectively. The grant stipulates that at least 25 temporary jobs must be created and at least 50 projects must be completed. It also requires that at least 20 lawn refurbishing jobs, one community garden, and three park cleanup jobs be completed. A survey of the town shows that at most 40 lawn refurbishing jobs are available, at most 4 community gardens are feasible, and there are 6 city parks. Formulate an integer program whose solution is a job selection strategy that minimizes cost to Greenside.

Solution. We let $x \in \mathbb{Z}^4$ be the number of each type of job to complete. In particular,

$$\begin{aligned} x_1 &= \text{number of lawns to refurbish,} \\ x_2 &= \text{number of trees to remove,} \\ x_3 &= \text{number of community gardens to develop,} \\ x_4 &= \text{number of parks to clean up.} \end{aligned}$$

The expected total cost, z, is the sum of the expected costs for each type of job.

$$z = 60x_1 + 200x_2 + 750x_3 + 250x_4.$$

At least 50 projects must be completed.

$$x_1 + x_2 + x_3 + x_4 \geq 50.$$

At least 25 temporary jobs must be created. Each lawn refurbishing job creates one-fifth of a temporary job, so the number of temporary jobs created by all lawn jobs is $x_1/5$. Using this reasoning we have the constraint:

$$\tfrac{1}{5}x_1 + \tfrac{3}{4}x_2 + 2x_3 + x_4 \geq 25.$$

Grant and city constraints require the additional restrictions:

$$\begin{aligned} 20 \leq x_1 &\leq 40, \\ 0 \leq x_2, \\ 1 \leq x_3 &\leq 4, \\ 3 \leq x_4 &\leq 6. \end{aligned}$$

Gathering our results, we have the IP:

$$\boxed{\begin{array}{llrrrrl}
\min & z = & 60x_1 & +200x_2 & +750x_3 & +250x_4 & \\
s.t. & & x_1 & +x_2 & +x_3 & +x_4 & \geq 50 \\
 & & \frac{1}{5}x_1 & +\frac{3}{4}x_2 & +2x_3 & +x_4 & \geq 25 \\
 & & \multicolumn{5}{l}{20 \leq x_1 \leq 40} \\
 & & \multicolumn{5}{l}{0 \leq x_2} \\
 & & \multicolumn{5}{l}{1 \leq x_3 \leq 4} \\
 & & \multicolumn{5}{l}{3 \leq x_4 \leq 6} \\
 & & \multicolumn{5}{l}{x \in \mathbb{Z}^4}
\end{array}}$$

The optimal solution is

$$\boxed{x^* = \begin{bmatrix} 29 & 15 & 1 & 6 \end{bmatrix}^T, \quad z^* = 6990.}$$

The optimal strategy for Greenside is to refurbish 29 lawns, remove 15 trees, create one community garden, and clean up all 6 city parks. The total cost to Greenside is \$6990. Hopefully, the government grant covers at least this cost. This strategy completes 51 projects and creates 25.05 temporary jobs.

Notice that the optimal solution is a scenario in which the number of temp jobs created is not integer. This does not violate any problem statement conditions. If the number of temp jobs needs to be further specified, then we would need a somewhat different IP formulation. To do this, create an additional decision variable $x_5 =$ (the number of temp jobs created) with $z_5 \in \mathbb{Z}$, and carefully reconsider the constraints.

4.3 Tourist Trap Shop

This example is a profit maximization problem with both revenue and costs. It illustrates consumer-demand and business-practice requirements.

The Tourist Trap Shop survives on the tourist trade selling a variety of trinkets. Among other items, they sell T-shirts, key chains, refrigerator magnets, and hats. They can obtain these items from four different wholesalers which supply these items in fixed quantities. The table below shows the quantity of each item found in each supplier's standard shipment. It also shows the wholesale cost of each shipment.

Supplier	Batch Cost ($)	T-shirts (#)	Key chains (#)	Magnets (#)	Hats (#)
A	172	60	26	32	26
B	60	24	0	0	12
C	96	0	20	20	14
D	48	10	0	10	10

In considering the tourist business forecast, the manager of Tourist Trap has decided to purchase at least 1000 of these items (in some combination) for the season. The total available purchasing funds are $2950. Key chains do not sell very well, so the total number of key chains should be limited to a maximum of 15% of the total number of items. T-shirts are always popular, so the total T-shirt count should be at least 40% of the total number of items. The total expected revenue per item for T-shirts, key chains, magnets and hats is $7, $2, $3, and $5, respectively. Formulate an integer program whose solution determines the best strategy for buying batches of items from the four suppliers with the goal of maximizing profit

Solution. We let $x \in \mathbb{Z}^4$ be the number of batches to purchase from each supplier. In particular, x_i is the number of batches to purchase from supplier i, where $i = 1, 2, 3, 4$, corresponding to A, B, C, D, respectively.

The expected profit is the total expected revenue less the cost of purchasing items. The expected revenue depends upon how many of each item is purchased as part of the various batches. For T-shirts, we have

$$\#\text{ of T-shirts } = 60x_1 + 24x_2 + 10x_4$$

and the total revenue for T-shirts is

$$\text{T-shirt revenue } = 7\left(60x_1 + 24x_2 + 10x_4\right).$$

Similarly, we have

$$\text{Key chain revenue } = 2\left(26x_1 + 20x_3\right),$$

$$\text{Magnet revenue } = 3\left(32x_1 + 20x_3 + 10x_4\right),$$
$$\text{Hat revenue } = 5\left(26x_1 + 12x_2 + 14x_3 + 10x_4\right).$$

The total profit, z, is the sum of these individual profits less the cost of batches:

$$z = 698x_1 + 228x_2 + 170x_3 + 150x_4 - 172x_1 - 60x_2 - 96x_3 - 48x_1,$$

$$z = 526x_1 + 168x_2 + 74x_3 + 102x_4.$$

Alternatively, one could compute the expected profit per batch from each supplier, arriving at the same profit function.

Next, consider the various constraints. First, have purchasing funds limited to \$2950. This leads to the constraint:

$$172x_1 + 60x_2 + 96x_3 + 48x_4 \leq 2950.$$

Next, we must purchase at least 1000 total items. Each batch contains different numbers of items. For example, we have

$$144x_1 + 36x_2 + 54x_3 + 30x_4 \geq 1000.$$

The number of key chains should be limited to a maximum of 15% of the total number of items. We have

$$26x_1 + 20x_3 \leq (0.15)(144x_1 + 36x_2 + 54x_3 + 30x_4).$$

Or, equivalently,

$$4.4x_1 - 5.4x_2 + 11.9x_3 - 4.5x_4 \leq 0.$$

Notice, in particular that the total number of purchased items is not necessarily equal to 1000. Similarly, we have that the total number of T-shirts should be at least 40% of the total number of items. This leads to the constraint:

$$2.4x_1 + 9.6x_2 - 21.6x_3 - 2x_4 \geq 0.$$

Thus, the IP:

$$\boxed{\begin{array}{llrrrrl}
\max & z = & 526x_1 & +168x_2 & +74x_3 & +102x_4 & \\
s.t. & & 172x_1 & +60x_2 & +96x_3 & +48x_4 & \leq 2950 \\
 & & 144x_1 & +36x_2 & +54x_3 & +30x_4 & \geq 1000 \\
 & & 4.4x_1 & -5.4x_2 & +11.9x_3 & -4.5x_4 & \leq 0 \\
 & & 2.4x_1 & +9.6x_2 & -21.6x_3 & -2x_4 & \geq 0 \\
 & & 0 \leq x & & & & \\
 & & x \in \mathbb{Z}^4 & & & &
\end{array}}$$

The optimal solution is

$$x^* = \begin{bmatrix} 13 & 11 & 0 & 1 \end{bmatrix}^T,$$

$$z^* = 8788.$$

This solution shows that Tourist Trap should purchase 13 batches from supplier 1, 11 batches from supplier 2 and one batch from supplier 4. The total expected profit is \$8788. With this solution, 2298 items are purchased, including 1054 T-shirts (45.9%) and 338 key chains (14.7%).

Notice that the number of items to be purchased far exceeds the minimum requirement. This suggests that the manager may want to reconsider the purchasing requirement as such a large number of items may effect expected sales/profit, or be indicative of some other reasonable constraint that was not considered in the modeling.

4.4 Woof&Meow Pet Food Company

This example illustrates the use of dependent variables to simplify problem presentation – at the expense of having more decision variables.

Woof&Meow has decided to use year-end overstock to produce specialty test-market canned food for cats and dogs. The food will be composed of five ingredients: chicken, beef, wheat, rice, and water. The following table shows, for each item, the quantity in stock, the fraction (by weight) of protein and fat, and the calorie content.

Ingredient	stock (g)	protein (wt frac)	fat (wt frac)	calories (per g)
chicken	31750	0.27	0.14	2.4
beef	21780	0.26	0.15	2.5
wheat	15600	0.14	0.024	3.4
rice	21000	0.027	0.0025	1.3
water	unlimited	0	0	0

Any food produced must meet certain nutritional requirements. Acceptable ranges are shown in the table below.

	protein range (wt frac)	fat range (wt frac)	calorie range (per g)
cat food	0.07–0.12	0.04–0.07	1.00–1.25
dog food	0.09–0.13	0.03–0.08	0.75–1.15

In addition, consumer appreciation and marketability requirement suggest that any food produced must contain at least 30% meat by weight. Formulate a linear program whose solution provides recipes for a batch of dog food and a batch of cat food that together maximize the total weight of overstock used in the process.

Solution. We must determine how much of each ingredient is used in each type of food. Let

$$x_{ij} = \text{ the grams of ingredient } i \text{ used in food } j,$$

where $i = 1, 2, 3, 4, 5$ corresponds to chicken, beef, wheat, rice and water, respectively; and $j = 1, 2$ corresponds to cat food and dog food, respectively. We can consider $x_{ij} \in \mathbb{R}$. Because there are many constraints on various relative quantities, we will also use the following dependent variables.

$$W_j = \text{ the total weight of food } j.$$

$$W_{pj} = \text{ the total protein weight in food } j.$$
$$W_{fj} = \text{ the total fat weight in food } j.$$
$$C_j = \text{ the total calories in food } j.$$

All together there are 18 decision variables: ten original variables and eight dependent variables.

The objective is to maximize the use of overstock ingredients (by weight). That is, we seek to maximize the objective function

$$z = x_{11} + x_{21} + x_{31} + x_{41} + x_{12} + x_{22} + x_{32} + x_{42}.$$

We begin enumerating the problem constraints by defining our dependent variables in terms of our initial decision variables.

Remark 14. *Whenever dependent variables are created for an optimization problem, the dependencies must be included in the constraint set.*

$$\begin{aligned}
W_1 &= x_{11} + x_{21} + x_{31} + x_{41} + x_{51}, \\
W_2 &= x_{12} + x_{22} + x_{32} + x_{42} + x_{52}, \\
W_{p1} &= 0.27x_{11} + 0.26x_{21} + 0.14x_{31} + 0.027x_{41}, \\
W_{f1} &= 0.14x_{11} + 0.15x_{21} + 0.024x_{31} + 0.0025x_{41}, \\
W_{p2} &= 0.27x_{12} + 0.26x_{22} + 0.14x_{32} + 0.027x_{42}, \\
W_{f2} &= 0.14x_{12} + 0.15x_{22} + 0.024x_{32} + 0.0025x_{42}, \\
C_1 &= 2.4x_{11} + 2.5x_{21} + 3.4x_{31} + 1.3x_{41}, \\
C_2 &= 2.4x_{12} + 2.5x_{22} + 3.4x_{32} + 1.3x_{42}.
\end{aligned}$$

Now, our inequality constraints are relatively simple to express in terms of our augmented set of decision variables. For example, the requirement that cat food be at least 30% meat by weight is expressed as

$$\frac{x_{11} + x_{21}}{W_1} \geq 0.3 \quad \text{or equivalently} \quad -x_{11} - x_{21} + 0.3W_1 \leq 0.$$

Notice that the above equivalence assumes that $W_1 > 0$. For this particular problem we can be reasonably assured that this will be the case at the optimal solution. That is, we expect $W_1^* > 0$. However, the denominator of any such constraint should be checked once the problem is solved to make sure that the equivalency is met.

Remark 15. *Suppose x and y are decision variables and b is a constant. A constraint such as $x/y \geq b$ is equivalent to $x \geq by$ only if $y^* > 0$.*

The corresponding constraint for dog food is

$$\frac{x_{12}+x_{22}}{W_2} \geq 0.3 \quad \text{or equivalently} \quad -x_{12}-x_{22}+0.3W_2 \leq 0.$$

There are constraints on the weight fraction of proteins. For the cat food we have

$$0.07 \leq \frac{W_{p1}}{W_1} \leq 0.12$$

which is equivalent to the two constraints

$$0.07W_1 - W_{p1} \leq 0 \quad \text{and} \quad -0.12W_1 + W_{p1} \leq 0.$$

For dog food we have the corresponding constraints

$$0.09W_2 - W_{p2} \leq 0 \quad \text{and} \quad -0.13W_2 + W_{p2} \leq 0.$$

We also have similar constraints on the weight fraction of fats:

$$\begin{aligned} 0.04W_1 - W_{f1} \leq 0 \quad &\text{and} \quad -0.07W_1 + W_{f1} \leq 0, \\ 0.03W_2 - W_{f2} \leq 0 \quad &\text{and} \quad -0.08W_2 + W_{f2} \leq 0. \end{aligned}$$

Next, there are restrictions on the calorie density for each food. Using the same type of construction, we have

$$\begin{aligned} W_1 - C_1 \leq 0 \quad &\text{and} \quad -1.25W_1 + C_1 \leq 0, \\ 0.75W_2 - C_2 \leq 0 \quad &\text{and} \quad -1.15W_2 + C_2 \leq 0. \end{aligned}$$

Finally, we cannot use more ingredients than are available:

$$\begin{aligned} x_{11}+x_{12} &\leq 31750, && \text{(chicken)} \\ x_{21}+x_{22} &\leq 21780, && \text{(beef)} \\ x_{31}+x_{32} &\leq 15600, && \text{(wheat)} \\ x_{41}+x_{42} &\leq 21000. && \text{(rice)} \end{aligned}$$

We now have the LP:

$$\boxed{\begin{array}{rl} \max_x & z = c^T x \\ \text{s.t.} & A_e = b_e \\ & A \leq b \\ & x \geq 0 \\ & x \in \mathbb{R}^{18} \end{array}}$$

where

$$x^T = \begin{bmatrix} x_{11} & x_{21} & x_{31} & x_{41} & x_{51} & x_{12} & x_{22} & x_{32} & x_{42} & x_{52} & W_1 & W_2 & W_{p1} & W_{f1} & W_{p2} & W_{f2} & C_1 & C_2 \end{bmatrix}$$

$$c^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A_e = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.27 & 0.26 & 0.14 & 0.027 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0.14 & 0.15 & 0.024 & 0.0025 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.27 & 0.26 & 0.14 & 0.027 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.14 & 0.15 & 0.024 & 0.0025 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 2.4 & 2.5 & 3.4 & 1.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2.4 & 2.5 & 3.4 & 1.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

$$b_e^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.07 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.12 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.09 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.13 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.04 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.07 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.03 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.08 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1.25 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.75 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1.15 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$b^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 31750 & 21780 & 15600 & 21000 \end{bmatrix}$$

The solution of this problem is left as an exercise.

4.5 Pack&Ship Delivery

This problem is an example of a packing problem. The formulation is presented in a general form with the use of parameter variables for quantities that are not decision variables. It also illustrates the use of summation notation to simplify the model presentation.

A transport ship has three compartments for storing cargo: front ($j = 1$), center ($j = 2$), rear ($j = 3$). Each compartment has limits on weight capacity, (w_j in tons) and space capacity (s_j in m^3). Any proportion of any of four cargo types ($i = 1, 2, 3, 4$) can be placed in these compartments for shipment. Each cargo type has total weight t_i (tons), volume density v_i (m^3/ton) and profit p_i (\$/ton). In order to balance the ship, the ratio of cargo weight to compartment capacity must be equal for all three compartments. How much of each cargo type should be distributed to each compartment in order to maximize profit?

Solution. We can choose how much of each cargo type is packed into each compartment. Let

$$x_{ij} = \text{ tons of cargo } i \text{ placed in compartment } j,$$

where $x_{i,j} \in \mathbb{R}$. The objective is to maximize profit. We can build the objective function as follows.

$$\begin{aligned}
z &= (\text{profit in \$}) \\
&= \sum_{i=1}^{4} (\text{profit in \$ from cargo type } i) \\
&= \sum_{i=1}^{4} (\text{profit per ton of cargo } i)\,(\text{tons of cargo } i \text{ shipped}) \\
&= \sum_{i=1}^{4} p_i(x_{i1} + x_{i2} + x_{i3}) \\
&= \sum_{i=1}^{4} p_i \sum_{j=1}^{3} x_{ij} \\
&= \sum_{i=1}^{4} \sum_{j=1}^{3} p_i x_{ij}
\end{aligned}$$

Remark 16. *The use of summation notation is often preferred in composing the objective function and any constraints. This method saves space and can express many large constraints efficiently.*

There are several constraints. First, each compartment j has a weight limit.

$$\text{(total weight packed in compartment } j) \leq \text{(weight limit of compartment } j)$$
$$x_{1j} + x_{2j} + x_{3j} + x_{4j} \leq w_j$$
$$\sum_{i=1}^{4} x_{ij} \leq w_j.$$

So, we have the set of constraints (one for each value of j):

$$\sum_{i=1}^{4} x_{ij} \leq w_j, \quad j = 1, 2, 3.$$

Also, wach compartment has a volume limit. This set of constraints uses the fact that each cargo type is described by a volume density, so $v_i x_{ij}$ is the volume of cargo type i packed in compartment j.

$$\sum_{i=1}^{4} v_i x_{ij} \leq s_j, \quad j = 1, 2, 3.$$

There is a limited quantity (by weight) of each type of cargo.

$$\sum_{j=1}^{3} x_{ij} \leq t_i, \quad i = 1, 2, 3, 4.$$

The ship must be properly balanced in that the ratio of packed weight to available volume must be equal for each compartment.

$$\frac{1}{s_1} \sum_{i=1}^{4} x_{i1} = \frac{1}{s_2} \sum_{i=1}^{4} x_{i2} = \frac{1}{s_3} \sum_{i=1}^{4} x_{i3}.$$

This set of equality constraints is equivalent to the two constraints:

$$s_2 \sum_{i=1}^{4} x_{i1} - s_1 \sum_{i=1}^{4} x_{i2} = 0,$$

$$s_3 \sum_{i=1}^{4} x_{i2} - s_2 \sum_{i=1}^{4} x_{i3} = 0,$$

Remark 17. *Notice that a constraint set $g_1(x) = g_2(x) = \cdots = g_m(x)$ is equivalent to the $m-1$ constraints $g_1(x) = g_2(x)$, $g_2(x) = g_3(x)$, $\ldots$ $g_{m-1}(x) = g_m(x)$. All other mixed constraint pairs $g_k(x) = g_j(x)$ are then automatically satisfied.*

Finally, we must specify that we cannot pack negative quantities of any cargo.

$$x \geq 0.$$

We next collect our work into a linear program with 12 decision variables, 12 constraints and 12 sign restrictions. The fact that there are 12 of each is entirely coincidental. We have

$$\boxed{\begin{array}{ll} \max_x & z = c^T x \\ \text{s.t.} & A_e x = b_e \\ & Ax \leq b \\ & x \geq 0 \\ & x \in \mathbb{R}^{12}, \end{array}}$$

where we specify the variable order

$$x^T = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{21} & x_{22} & x_{23} & x_{31} & x_{32} & x_{33} & x_{41} & x_{42} & x_{43} \end{bmatrix},$$

and the other matrices:

$$c^T = \begin{bmatrix} p_1 & p_1 & p_1 & p_2 & p_2 & p_2 & p_3 & p_3 & p_3 & p_4 & p_4 & p_4 \end{bmatrix}$$

$$A_e = \begin{bmatrix} s_2 & -s_1 & 0 & s_2 & -s_1 & 0 & s_2 & -s_1 & 0 & s_2 & -s_1 & 0 \\ 0 & s_3 & -s_2 & 0 & s_3 & -s_2 & 0 & s_3 & -s_2 & 0 & s_3 & -s_2 \end{bmatrix}$$

$$b_e^T = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ v_1 & 0 & 0 & v_2 & 0 & 0 & v_3 & 0 & 0 & v_4 & 0 & 0 \\ 0 & v_1 & 0 & 0 & v_2 & 0 & 0 & v_3 & 0 & 0 & v_4 & 0 \\ 0 & 0 & v_1 & 0 & 0 & v_2 & 0 & 0 & v_3 & 0 & 0 & v_4 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$b^T = \begin{bmatrix} w_1 & w_2 & w_3 & s_1 & s_2 & s_3 & t_1 & t_2 & t_3 & t_4 \end{bmatrix}$$

The solution to this problem will depend on the parameterized quantities w, s, t, p, and v.

4.6 Shoes All Year

This modeling problem is an example of multi-period production planning. We also illustrate one method for linearizing objective functions that contain absolute value terms.

A shoe company needs to set up a monthly production schedule for the year that meets predicted monthly demand d_k, $k = 1, 2, \ldots, 12$. Surplus shoes at the end of any month must be stored at a cost of t dollars per pair. Changing production quantity from one month to the next incurs a *positive* factory retooling cost of r dollars per pair (increase or decrease). Suppose that the company begins the year with no stored shoes and wishes to end the year with no surplus shoes. Construct an integer program that can be used to minimize production and storage costs while meeting monthly demand.

Solution. The company might choose to make all required shoes $\sum_{k=1}^{12} d_k$ in the first month, but this would result in high storage costs as shoes are held for future months in the year. The company might also choose to produce just enough shoes each month to meet the demand for that month. However, this plan might result in large retooling costs as production is changed month to month. The optimal solution will find a balance to these conflicting costs. Let

$$\begin{aligned} x_k &= \text{ (shoe production volume in month } k = 1, 2, \ldots, 12), \\ s_k &= \text{ (surplus volume at the end of month } k = 1, 2, \ldots, 11). \end{aligned}$$

Notice that the surplus variables are dependent variables and need to be expressed in terms of the monthly production volumes and demands. In general, the surplus at the end of any month is equal to the surplus from the previous month plus the production in that month minus the demand for that month:

$$s_k = s_{k-1} + x_k - d_k.$$

The first and last months are somewhat different because of the surplus conditions stated in the problem. We have

$$\begin{aligned} s_1 &= 0 + x_1 - d_1 && \text{(month 1)} \\ s_2 &= s_1 + x_2 - d_2 && \text{(month 2)} \\ s_3 &= s_2 + x_3 - d_3 && \text{(month 3)} \\ &\vdots \\ s_{11} &= s_{10} + x_{11} - d_{11} && \text{(month 11)} \\ 0 &= s_{11} + x_{12} - d_{12} && \text{(month 12)} \end{aligned}$$

The objective function measures the retooling and storage costs which we construct in three terms:

$$z = rx_1 + r\sum_{k=2}^{12} |x_k - x_{k-1}| + t\sum_{k=1}^{11} s_k.$$

The first term is the initial retooling cost of r dollars per pair. The second term includes the retooling costs for the next 11 months. For each month, the retooling depends upon the absolute change in monthly production. The last term includes the storage costs for each of the first 11 months.

This objective function is nonlinear, and we will address this issue shortly. We must also include constraints that make sure that we meet monthly demand. This demand can be met by requiring that the storage and production quantities are nonnegative:

$$x \geq 0, \quad s \geq 0.$$

There are no stated constraints on production volumes or resource limitations, though these type of constraints are typically present. The optimization problem is

$$\boxed{\begin{array}{ll} \min\limits_{x,s} & z = rx_1 + r\sum\limits_{k=2}^{12} |x_k - x_{k-1}| + t\sum\limits_{k=1}^{11} s_k \\ \text{s.t.} & s_1 = 0 + x_1 - d_1 \\ & s_k = s_{k-1} + x_k - d_k, \qquad k = 2, 3, \ldots, 11 \\ & 0 = s_{11} + x_{12} - d_{12} \\ & x, s \geq 0 \\ & x \in \mathbb{Z}^{12}, s \in \mathbb{Z}^{11}. \end{array}}$$

The absolute value terms in the objective function are nonlinear. Fortunately, we can reformulate this optimization problem into an equivalent integer program. The retooling costs terms do not necessarily satisfy $x_k - x_{k-1} \geq 0$. For these 11 terms, we introduce corresponding excess variables y_k and deficit variables w_k defined by

$$y_k = \begin{cases} x_k - x_{k-1} & \text{if } x_k \geq x_{k-1} \\ 0 & \text{if } x_k < x_{k-1} \end{cases}$$

$$w_k = \begin{cases} 0 & \text{if } x_k \geq x_{k-1} \\ x_{k-1} - x_k & \text{if } x_k < x_{k-1} \end{cases}$$

Observe that if $x_k \geq x_{k-1}$ then y_k is the (positive) change and $w_k = 0$. Also, if $x_k < x_{k-1}$ then w_k is the (positive) change and $y_k = 0$. With these new variables,

notice the following two relations.

$$y_k - w_k = \begin{cases} x_k - x_{k-1} & \text{if } x_k \geq x_{k-1} \\ x_k - x_{k-1} & \text{if } x_k < x_{k-1} \end{cases} = x_k - x_{k-1},$$

$$y_k + w_k = \begin{cases} x_k - x_{k-1} & \text{if } x_k \geq x_{k-1} \\ x_{k-1} - x_k & \text{if } x_k < x_{k-1} \end{cases} = |x_k - x_{k-1}|.$$

We can use the second set of relations to create a linear objective function. The first set of relations becomes a new set of constraints that describe the relationship between the original (independent) variables and the new (dependent) variables. The equivalent integer program for this problem is

$$\boxed{\begin{array}{lll} \min\limits_{x,s,y,w} & z = rx_1 + r\sum\limits_{k=2}^{12}(y_k + w_k) + t\sum\limits_{k=1}^{11} s_k & \\ \text{s.t.} & s_1 = 0 + x_1 - d_1 & \\ & s_k = s_{k-1} + x_k - d_k, & k = 2, 3, \ldots, 11 \\ & 0 = s_{11} + x_{12} - d_{12} & \\ & y_k - w_k = x_k - x_{k-1}, & k = 2, 3, \ldots, 11 \\ & x, s, y, w \geq 0 & \\ & x \in \mathbb{Z}^{12}; \quad s, y, w \in \mathbb{Z}^{11}. & \end{array}}$$

Remark 18. *The following two minimization problems are equivalent.*

$$\begin{array}{ll} \min\limits_{x} & z = |x| \\ s.t. & Ax \leq b \\ & x \in \mathbb{R} \end{array} \qquad \begin{array}{ll} \min\limits_{x,y,w} & z = y + w \\ s.t. & y - w = x \\ & Ax \leq b \\ & y, w \geq 0 \\ & x, y, w \in \mathbb{R} \end{array}$$

Clearly, the two given optimization problems in Remark 18 are not identical. We say that they are *equivalent* because the optimal solution to one problem provides the optimal solution to the other problem. In the present example, solving the linear program yields x^*, y^*, w^*, z^*. Then, the solution to the nonlinear problem is x^*, z^*.

4.7 Thermo – (Linear Data Fitting)

This example demonstrates how linear programming (and least squares methods) can be used to find an affine function that minimizes error between dependent and independent data values. We also illustrate a second method for linearizing absolute value terms in the objective function. This example is also accompanied by Octave *and* Python *code.*

Thermo manufactures rapid-response partial immersion thermometers for use in laboratory settings. For each thermometer, the company produces a set of calibration temperature measurements and supplies these to the customer. They would also like to provide the best linear fit to the data so that the customer has the option to automate the use of the thermometer in a more dynamic setting that does not require constant user interpretation. Consider the set of calibration data shown in Table 4.1.

k	1	2	3	4	5	6	7	8	9
measured temp, x_k ($^\circ C$)	10.4	21.0	29.8	39.7	50.4	60.0	71.2	87.2	90.3
actual temp, y_k ($^\circ C$)	9.5	20.1	29.5	38.9	50.0	59.2	71.0	81.8	89.9

Table 4.1: Calibration data for a single thermometer manufactured by Thermo.

Thermo would like to produce a best linear fit to this data $y = ax + b$ so that a measured temperature x is quickly converted to a temperature y which is close to the actual temperature. That is, given the calibration data, what values of a and b provide the best linear fit $y = ax + b$?

Before we answer this question, it is interesting to look at the data set. The eighth measured temperature, x_8, appears to be anomalously high, perhaps by as much as five degrees. We will not try to fix this apparent error in reading (or in transcription), nor will we remove this datum from the data set. We will consider two methods of data fitting and compare their behavior with respect to this data set as a whole.

Solution #1. We can solve the Thermo problem using the method of least squares. Using our knowledge of optimization, we can formulate the relevant optimization problem and solve it without resorting to looking up formulas from other sources. In other words, we can truly understand *why* the least squares formulas work.

We seek coefficients a and b so that $y_k \approx ax_k + b$ for all k. One measure of goodness is to minimize the sum of the squared deviations:

$$z = \sum_{k=1}^{9} (ax_k + b - y_k)^2,$$

where $ax_k + b$ is the temperature predicted by the model and y_k is the actual temperature. The above objective computes all differences between prediction and actual,

squares them, then adds them together. If it is possible to choose a and b so that predicted and actual temperatures are identical, then $z = 0$, and otherwise, $z > 0$. This optimization problem has no constraints on a and b. We have

$$\boxed{\begin{array}{ll} \min\limits_{a,b} & z = \sum\limits_{k=1}^{9} (ax_k + b - y_k)^2 \\ \text{s.t.} & a, b \in \mathbb{R}. \end{array}}$$

This is an unconstrained optimization problem in two variables: a and b. Thus, any optimal points will be stationary points of the objective function. First, compute the gradient of the objective with respect to the decision variables.

$$\begin{aligned}
\nabla z(a,b) &= \begin{bmatrix} \frac{\partial z}{\partial a} \\ \frac{\partial z}{\partial b} \end{bmatrix} \\
&= \begin{bmatrix} 2\sum\limits_{k=1}^{9} x_k (ax_k + b - y_k) \\ 2\sum\limits_{k=1}^{9} (ax_k + b - y_k) \end{bmatrix} \\
&= 2 \begin{bmatrix} a\sum\limits_{k=1}^{9} x_k^2 + b\sum\limits_{k=1}^{9} x_k - \sum\limits_{k=1}^{9} x_k y_k \\ a\sum\limits_{k=1}^{9} x_k + b\sum\limits_{k=1}^{9} 1 - \sum\limits_{k=1}^{9} y_k \end{bmatrix} \\
&= 2 \begin{bmatrix} \sum\limits_{k=1}^{9} x_k^2 & \sum\limits_{k=1}^{9} x_k \\ \sum\limits_{k=1}^{9} x_k & \sum\limits_{k=1}^{9} 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} - 2 \begin{bmatrix} \sum\limits_{k=1}^{9} x_k y_k \\ \sum\limits_{k=1}^{9} y_k \end{bmatrix}
\end{aligned}$$

Stationary points satisfy $\nabla z(a,b) = 0$, which leads to

$$\begin{bmatrix} a^* \\ b^* \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^{9} x_k^2 & \sum_{k=1}^{9} x_k \\ \sum_{k=1}^{9} x_k & \sum_{k=1}^{9} 1 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{k=1}^{9} x_k y_k \\ \sum_{k=1}^{9} y_k \end{bmatrix}$$

$$= \begin{bmatrix} 29980.82 & 460 \\ 460 & 9 \end{bmatrix}^{-1} \begin{bmatrix} 29322.46 \\ 449.9 \end{bmatrix}$$

$$\approx \begin{bmatrix} 0.9780 \\ 0.0007 \end{bmatrix}.$$

This solution represents the linear fit function $y = (0.9780)x + (0.0007)$. We leave it to the reader to show that this stationary point is a global minimizer of the objective function. Both the data and best fit line are shown in Figure 4.1.

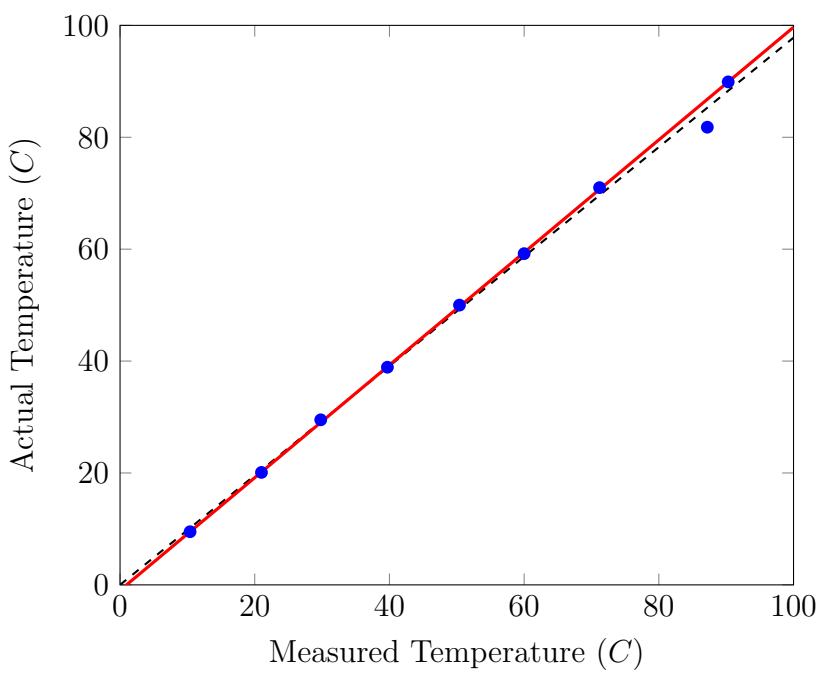


Figure 4.1: Raw data (blue circles), least squares linear fit (black dashed line) and least deviation linear fit (red solid line) for the Thermo calibration data.

Solution #2. Next, we solve the problem by the method of least deviation using the same model fitting function $y = ax + b$. In this case, we consider an objective function which measures the least total deviation between the prediction and the data, rather than the square. We have

$$z = \sum_{k=1}^{9} |ax_k + b - y_k|.$$

This objective function could be linearized with the use of excess and deficit variables as was done with the Shoes All Year problem. We will introduce a second method which requires fewer new decision variables, but more additional constraints. Notice that for any real value w, $|w| = \max\{w, -w\}$.

Remark 19. *The above discussion suggests that these two optimization problems are equivalent:*

$$\begin{aligned} \min_{w} \quad & z = |w| \\ s.t. \quad & Aw \leq b \\ & w \in \mathbb{R} \end{aligned}$$

$$\begin{aligned} \min_{w,\delta} \quad & z = \delta \\ s.t. \quad & \delta \geq w \\ & \delta \geq -w \\ & Aw \leq b \\ & w \in \mathbb{R}. \end{aligned}$$

These linear programs return the same z^* and w^* though they have different feasible regions. Notice that whenever $\delta > |w|$, the LP can choose a smaller δ so that either $\delta = w$ or $\delta = -w$. That is, $\delta^* = \max\{w^*, -w^*\} = |w^*|$.

So, we have a linearization of the least deviation problem:

$$\begin{aligned} \min_{a,b,\delta} \quad & z = \sum_{k=1}^{9} \delta_k \\ s.t. \quad & \delta_k \geq ax_k + b - y_k \qquad k = 1, 2, \ldots, 9 \\ & \delta_k \geq -ax_k - b + y_k \qquad k = 1, 2, \ldots, 9 \\ & a, b \in \mathbb{R} \\ & \delta \in \mathbb{R}^9. \end{aligned}$$

At the optimal solution a^* and b^* are the optimal model coefficients and the δ_k^* are the absolute deviations between the model prediction and each data point. For the given calibration data, we have $(a^*, b^*) = (1.0063, -.9651)$. The best fit line is shown in Figure 4.1 as the red solid line.

The optimization model can be written somewhat compactly. Suppose that we have n data points (x_k, y_k). We can write the optimization problem as

$$
\begin{array}{lll}
\min\limits_{a,b,\delta} & z = 0a + 0b + \delta_1 + \delta_2 + \ldots + \delta_n & \\
s.t. & -x_1 a - b + \delta_1 & \geq -y_1 \\
& -x_2 a - b \qquad + \delta_2 & \geq -y_2 \\
& \qquad\qquad\qquad \vdots & \\
& -x_n a - b \qquad\qquad + \delta_n & \geq -y_n \\
& +x_1 a + b + \delta_1 & \geq +y_1 \\
& +x_2 a + b \qquad + \delta_2 & \geq +y_2 \\
& \qquad\qquad\qquad \vdots & \\
& +x_n a + b \qquad\qquad + \delta_n & \geq +y_n \\
& a, b \in \mathbb{R} & \\
& \delta \in \mathbb{R}^n &
\end{array}
$$

The symmetry of the constraints and simplicity of the objective allow us to construct a very compact representation for the problem. First, consider the objective function, which does not depend on either a or b and depends on the δ_k with unit coefficients. If we collect our decision variables in the ordered vector

$$
w^\mathsf{T} = \begin{bmatrix} a & b & \delta_1 & \delta_2 & \ldots & \delta_n \end{bmatrix},
$$

then we can write

$$
z = \begin{bmatrix} 0 & 0 & 1 & 1 & \ldots & 1 \end{bmatrix} w = \begin{bmatrix} 0 & 0 & \mathbf{1}_n^\mathsf{T} \end{bmatrix} w,
$$

where $\mathbf{1}_n$ is a vector of n ones. Next, consider the first n inequality constraints. The variable a is multiplied by the negative of each x value, the variable b is multiplied by negative one, and each δ shows up only in one position in each constraint. The coefficient matrix is

$$
\begin{bmatrix} -x & -\mathbf{1}_n & I_n \end{bmatrix},
$$

and the full set of constraints is

$$
\begin{bmatrix} -x & -\mathbf{1}_n & I_n \end{bmatrix} \begin{bmatrix} a \\ b \\ \delta \end{bmatrix} \geq \begin{bmatrix} -y \end{bmatrix}.
$$

The remaining constraints can be modeled similarly. The resulting representation is

$$\boxed{\begin{aligned} \min_{w} \quad & z = c^T w \\ \text{s.t.} \quad & Aw \geq B \\ & w \in \mathbb{R}^{n+2} \\ & w^T = \begin{bmatrix} a & b & \delta^{\mathsf{T}} \end{bmatrix} \\ & c^T = \begin{bmatrix} 0 & 0 & \mathbf{1}_n^{\mathsf{T}} \end{bmatrix} \\ & A = \begin{bmatrix} -x & -\mathbf{1_n} & I_n \\ +x & +\mathbf{1_n} & I_n \end{bmatrix} \\ & B = \begin{bmatrix} -y \\ +y \end{bmatrix}. \end{aligned}} \tag{4.1}$$

Remember that the x and y values are given data, not problem variables.

Remark 20. *The matrix A above is written in <u>block matrix</u> form where entire rectangular sub-arrays are expressed as a single variable. This method saves space and can greatly facilitate the use of software tools.*

Discussion. Notice that the least deviation fit is less influenced by the presence of the outlier point (x_8, y_8). We may be more inclined to accept the least deviation result because it more closely approximates eight of the nine data points. A detailed description is beyond the scope of this work; however, we can make a general observation. The least squares method is appropriate if the uncertainty in the data is stationary random Gaussian. The least deviation method may be desirable if the data contains some outliers whose cause is of unknown origin. In this case, we might wonder if the 87.2 reading was actually 81.2 but was copied to the data set incorrectly.

Octave Solution to Linear Least Deviation. We can use the optimization problem (4.1) to illustrate how to quickly construct large matrices relevant to optimization problems. The following code solves the linear least deviation problem in Octave . It makes use of the block matrix structure of the various matrices and vectors.

```
x=[10.4;21.0;29.8;39.7;50.4;60.0;71.2;87.2;90.3];
y=[ 9.5;20.1;29.5;38.9;50.0;59.2;71.0;81.8;89.9];
N=length(x);
u=ones(N,1);
c=[0;0;u];
A=[-x -u eye(N) ; x u eye(N) ];
b=[-y ; y];
lb=[-inf ; -inf ; zeros(N,1)];
ub=inf(N+2,1);
ct=repmat("L",1,2*N);
vt=repmat("C",1,N+2);
sense=1;
[xstar,zstar]=glpk(c,A,b,lb,ub,ct,vt,sense);
```

The code makes use of several built-in Matlab /Octave functions. Here is a list that includes these examples and some additional useful commands.

- ones(r,c) creates a matrix (or vector) with r rows and c columns where each entry is 1. ones(2,4) returns

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

- zeros(r,c) creates a matrix (or vector) with r rows and c columns where each entry is 0. zeros(1,3) returns

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}.$$

- inf(r,c) creates a matrix (or vector) with r rows and c columns where each entry is ∞. inf(3,1) returns

$$\begin{bmatrix} \text{inf} \\ \text{inf} \\ \text{inf} \end{bmatrix}.$$

- eye(n) creates the $n \times n$ identity matrix. eye(3) returns

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- repmat(A,r,c) creates a block matrix with matrix A repeated r times vertically and c times horizontally. repmat([1 2 3],2,1) returns

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}.$$

- repmat(str,1,c) creates a string variable with string entry str repeated c times. repmat('hwo',1,3) returns 'hwohwohwo'.

- kron(T,S) creates a block matrix with block entry $B_{ij} = T_i S$. That is every entry T_{ij} in template T is replaced by matrix S multiplied by T_{ij}. Examples:

1. $\mathsf{T} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$, $\mathsf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\mathsf{kron(T,S)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$.

2. $\mathsf{T} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$, $\mathsf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\mathsf{kron(T,S)} = \begin{bmatrix} 1 & 0 & 2 & 0 & 3 & 0 \\ 0 & 1 & 0 & 2 & 0 & 3 \end{bmatrix}$.

3. $\mathsf{T{=}eye(3)}$, $\mathsf{S} = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $\mathsf{kron(T,S)} = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$.

4. $\mathsf{T} = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $\mathsf{S{=}eye(3)}$, $\mathsf{kron(T,S)} = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2 \end{bmatrix}$.

The following code solves the linear least deviation problem in Python. Notice that the inequality constraints must be modified to be upper bound constraints. That is, we use $-Aw \leq -B$ instead of $Aw \geq B$.

```
import numpy as np
import scipy.optimize as opt
x=[ 10.4 , 21.0 , 29.8 , 39.7 , 50.4 , 60.0 , 71.2 , 87.2 , 90.3 ]
y=[  9.5 , 20.1 , 29.5 , 38.9 , 50.0 , 59.2 , 71.0 , 81.8 , 89.9 ]
N=len(x)
u=np.ones((N,1))
v=np.zeros((2,1))
E=np.eye(N)
c=np.concatenate((v,u),axis=0)
xc=np.array(x).reshape((N,1))
yc=np.array(y).reshape((N,1))
A1=np.concatenate((-xc,-u,E),axis=1)
A2=np.concatenate((xc,u,E),axis=1)
A=np.concatenate((A1,A2),axis=0)
b=np.concatenate((-yc,yc),axis=0)
bds=(((-np.inf,np.inf), )*(N+2))
res=opt.linprog(c,-A,-b,bounds=bds)
```

We have used several functions and methods to achieve this result. The `numpy` functions `ones`, `zeros` and `eye` are identical in concept with the Octave functions of the same name, producing patterned arrays of ones and zeros. The function `concatenate` combines two or more `numpy` arrays by stacking vertically ($axis = 0$) or horizontally ($axis = 1$). We have also used a Python instance `np.inf` which is identical to Octave `inf`. The method `reshape` recasts an array into that with the same elements but of a different shape, similarly to the Octave `reshape` function. Finally, notice that in calling the `opt.linprog` function, both `Ae` and `be` have been omitted indicating that no equality constraints are present in this problem.

4.8 Atmospheric CO_2 – Nonlinear Data Fitting

This example demonstrates that least-deviation methods can also be used to model data using a subclass of nonlinear functions. Demonstration Octave *and* Python *code is provided.*

It may come as a surprise that we can use linear programming to solve many nonlinear data fitting problems. The key aspect is that the model of the data is expressible as a linear combination of non-parameterized – but otherwise arbitrary – functions. Consider the monthly mean atmospheric carbon dioxide (CO_2) measurement data taken at Mauna Loa in Hawaii[1]. Data for several years is shown in Figure 4.2.

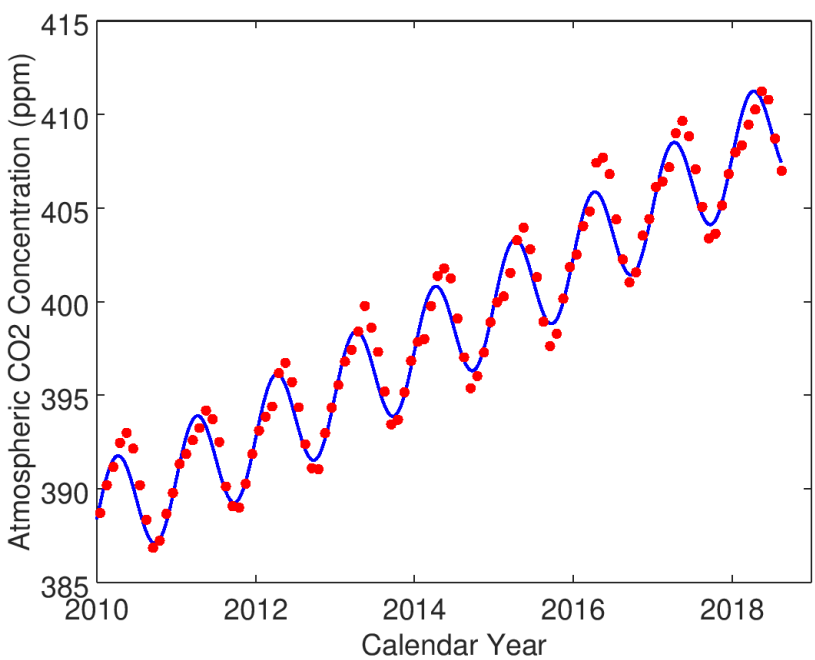


Figure 4.2: Monthly average CO_2 concentration as measured in the atmosphere on Mauna Loa (red) and a five-parameter least deviation fit to the data (blue).

The data suggests two major time-dependent features. First, there is a general upward trend that appears roughly linear. Second, there is a cyclic annual variation. A possible model that captures these features is

$$y = a_0 + a_1 x + a_2 x^2 + a_3 \sin(2\pi x) + a_4 \cos(2\pi x) .$$

[1]Source: NOAA Earth Systems Research Laboratory:
ftp://aftp.cmdl.noaa.gov/products/trends/co2/co2_mm_mlo.txt

Here, $x = T - 2010$ is the number of years since the beginning of 2010, where T is the calendar year (not necessarily integer) and y is the CO_2 concentration in parts per million. The coefficients a_0, a_1, a_2, a_3, a_4 are the undetermined parameters of the model – our decision variables. This model allows for a quadratic trend and annual cyclic variation. Both sine and cosine terms are included, with a period of one year, in order to capture an unknown phase shift in the sinusoidal function. In other words, our sinusoidal terms formulation is equivalent to the function $A\sin(2\pi x+\varphi)$ in the two parameters A and φ. The key element of this model is that it is a linear combination $y = \sum_{j=0}^{4} a_j\phi_j(x)$ of non-parameterized functions $\phi_0(x) = 1$, $\phi_1(x) = x$, $\phi_2(x) = x^2$, $\phi_3(x) = \sin(2\pi x)$, $\phi_4(x) = \cos(2\pi x)$.

We want to choose coefficients $a \in \mathbb{R}^5$ so that

$$y_k \approx a_0 + a_1x_k + a_2x_k^2 + a_3\sin(2\pi x_k) + a_4\cos(2\pi x_k)$$

for all data points (x_k, y_k). Using the method of least deviation, we have the optimization problem

$$\min_{a\in\mathbb{R}^5} \quad z = \sum_{k=1}^{N} \left|a_0 + a_1x_k + a_2x_k^2 + a_3\sin(2\pi x_k) + a_4\cos(2\pi x_k) - y_k\right|$$

and the equivalent linear formulation

$$\begin{aligned}
\min_{a,\delta} \quad & z = \sum_{k=1}^{N} \delta_k \\
s.t. \quad & \delta_k \geq +a_0 + a_1x_k + a_2x_k^2 + a_3\sin(2\pi x_k) + a_4\cos(2\pi x_k) - y_k \qquad k = 1, 2, \ldots, N \\
& \delta_k \geq -a_0 - a_1x_k - a_2x_k^2 - a_3\sin(2\pi x_k) - a_4\cos(2\pi x_k) + y_k \qquad k = 1, 2, \ldots, N. \\
& a \in \mathbb{R}^5 \\
& \delta \in \mathbb{R}^N
\end{aligned}$$

Upon some rearrangement of terms, we have

$$\begin{aligned}
\min_{a,\delta} \quad & z = \sum_{k=1}^{N} \delta_k \\
s.t. \quad & -a_0 - a_1x_k - a_2x_k^2 - a_3\sin(2\pi x_k) - a_4\cos(2\pi x_k) + \delta_k \geq -y_k \qquad k = 1, 2, \ldots, N \\
& +a_0 + a_1x_k + a_2x_k^2 + a_3\sin(2\pi x_k) + a_4\cos(2\pi x_k) + \delta_k \geq +y_k \qquad k = 1, 2, \ldots, N. \\
& a \in \mathbb{R}^5 \\
& \delta \in \mathbb{R}^N
\end{aligned}$$

which we can write in matrix form

$$
\begin{aligned}
\min_{w} \quad & z = c^T w \\
s.t. \quad & Aw \geq B \\
& w \geq 0 \\
& w \in \mathbb{R}^{N+5}
\end{aligned}
$$

$$
w^T = \begin{bmatrix} a_0 & a_1 & a_2 & a_3 & a_4 & \delta_1 & \delta_2 & \dots & \delta_N \end{bmatrix}
$$

$$
c^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 1 & \dots & 1 \end{bmatrix}
$$

$$
A = \begin{bmatrix} -\mathbf{1_N} & -x & -X & -S & -C & I_N \\ +\mathbf{1_N} & +x & +X & +S & +C & I_N \end{bmatrix}
$$

$$
B = \begin{bmatrix} -y \\ +y \end{bmatrix}
$$

where $\mathbf{1_N}$ is the column vector of N ones, I_N is the $N \times N$ identity matrix, X is the column vector of squared x values, S and C are the column vectors of values $\sin(2\pi x)$ and $\cos(2\pi x)$, respectively. The following Octave code solves the CO_2 data fitting problem (assuming data vectors x and y are defined).

```
N=length(x);
c=[zeros(5,1);ones(N,1)];
w=ones(N,1);
X=x.^2;
S=sin(2*pi*x);
C=cos(2*pi*x);
A=[-w -x -X -S -C eye(N) ; w x X S C eye(N)];
b=[-y ; y];
lb=-inf(N+5,1);
ub=[];
ct=repmat("L",1,2*N);
vt=repmat("C",1,N+5);
sense=1;
[xstar,zstar]=glpk(c,A,b,lb,ub,ct,vt,sense);
```

The following Python code solves the $C0_2$ data fitting problem. The x and y data are assumed to already exist as $N \times 1$ numpy arrays.

```
import numpy as np
import scipy.optimize as opt
N=len(x)
w=np.ones((N,1))
X=x**2
S=np.sin(2*np.pi*x)
C=np.cos(2*np.pi*x)
c=np.concatenate((np.zeros((5,1)),w),axis=0)
A1=np.concatenate(( w, x, X, S, C,-np.eye(N)),axis=1)
A2=np.concatenate((-w,-x,-X,-S,-C,-np.eye(N)),axis=1)
A=np.concatenate((A1,A2),axis=0)
b=np.concatenate((y,-y),axis=0)
bds=(((-np.inf,np.inf), )*(N+5))
res=opt.linprog(c,A,b,bounds=bds)
```

The optimal model parameters are

$$a^* = \left[\begin{array}{ccccc} 388.6 & 2.114 & 0.0314 & 2.976 & -0.840 \end{array}\right]^T .$$

We can interpret these values as follows.

- The seasonally adjusted CO_2 level at the beginning of year 2010 was 389 ppm.
- The annual increase in CO_2 level is 2.11 ppm at the beginning of year 2010.
- The annual increase in CO_2 level is 2.36 ppm at year 2018 ($a_1 + 2a_2(8) = 2.36$).
- The phase of annual variation is close to zero, most of the amplitude being in the sine term.
- The seasonal variation in CO_2 concentration has amplitude 3.09 ppm ($3.09 \approx \sqrt{(2.976)^2 + (0.840)^2}$).

4.9 ExaByte Networking

This example illustrates how linear programming can answer questions about paths on weighted graphs. Often such problems are difficult to model efficiently. The utility of box constraints is emphasized.

The ExaByte company specializes in rapid data transfer systems. They are considering using the existing computer network shown in Figure 4.3 for daily operations. Each network node (labeled with blue lower case roman letters) is connected to two or more other nodes. Each connection is indicated by a red line. The red numbers indicate the maximum operating transfer rate (bandwidth) for each connection. ExaByte would like to know the maximum transfer rate achievable between starting node s and target node t, and what routing should data take within the network. Data does not need to take a single path; it can be divided and routed among any combination of paths.

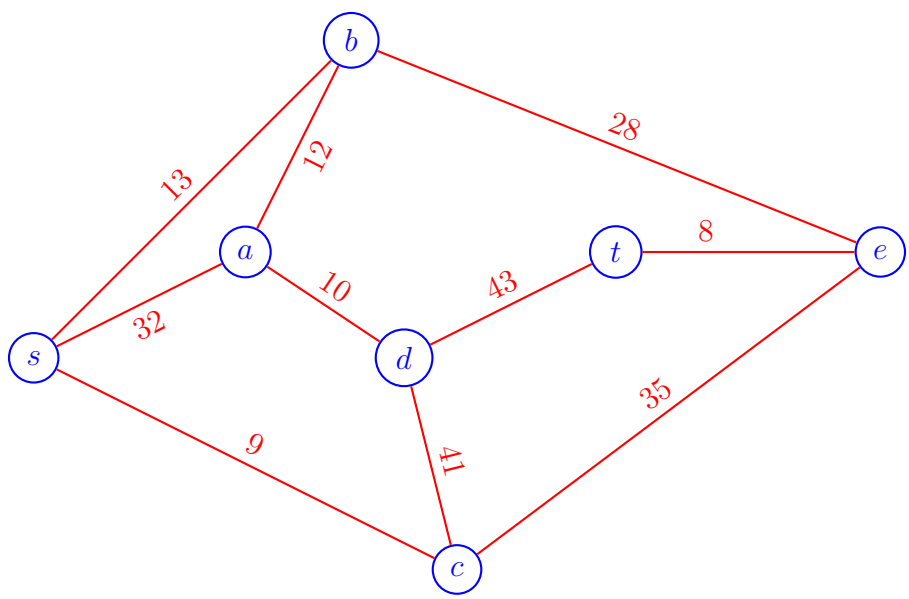


Figure 4.3: Representation of a 7-node computer network. Roman letters indicate computers, red lines indicate hardwired connections, and red numbers indicate maximum data transfer rates (bandwidths) along each connection.

Solution. We will assume that there is no data storage at any node, so that data only passes through the intermediate nodes a, b, c, d and e. Also, data can travel either direction along any connection. The maximum possible transfer rate out of the starting node is $32+13+9=54$. The maximum possible transfer rate into the target node is $43+8=51$. So, whatever the optimal transfer rate is, it will not be larger

than 51. It remains to be seen whether or not the network can support an overall transfer rate of 51.

We choose decision variables which measure the data transfer rate used along each network connection. Let

$$x_{ij} = (\text{utilized transfer rate from node } i \text{ to node } j),$$

where i and j are taken from the node set $\{s, a, b, c, d, e, t\}$. In particular, we have the 10 decision variables

$$x = \begin{bmatrix} x_{sa} & x_{sb} & x_{sc} & x_{ab} & x_{ad} & x_{be} & x_{cd} & x_{ce} & x_{dt} & x_{et} \end{bmatrix}^T.$$

Each decision variable can take on positive or negative values with the meaning:

$$x_{ij} > 0 \Rightarrow \text{data is moving from } i \text{ to } j,$$
$$x_{ij} < 0 \Rightarrow \text{data is moving from } j \text{ to } i.$$

The objective is to maximize the transfer rate from node s to node t. But as we have already seen, this can be measured by maximizing the transfer rate either out of node s or into node t. If we choose the latter, we have

$$z = x_{dt} + x_{et}.$$

Flow along each connection is limited by the connection bandwidth. For example, the connection between node a and node b has a bandwidth of 12, so we have the constraints

$$-12 \leq x_{ab} \leq 12.$$

Since we need not consider data transfer into the starting node we have

$$0 \leq x_{sa} \leq 32.$$

And, since we need not consider data transfer out of the target node we have

$$0 \leq x_{dt} \leq 43.$$

Taken all together, we have the box constraints on bandwidth

$$\begin{aligned}
0 &\leq x_{sa} \leq 32 \\
0 &\leq x_{sb} \leq 13 \\
0 &\leq x_{sc} \leq 9 \\
-12 &\leq x_{ab} \leq 12 \\
-10 &\leq x_{ad} \leq 10 \\
-28 &\leq x_{be} \leq 28 \\
-41 &\leq x_{cd} \leq 41 \\
-35 &\leq x_{ce} \leq 35 \\
0 &\leq x_{dt} \leq 43 \\
0 &\leq x_{et} \leq 8
\end{aligned}$$

We also must consider that no data storage can occur at intermediate nodes. This means, for example, that the total data rate into node a must be equal to the total data rate out of node a. This is represented by the constraint

$$\begin{aligned}(\text{data rate into node } a) &= (\text{data rate out of node } a)\\ x_{sa} &= x_{ab} + x_{ad}.\end{aligned}$$

This constraint holds whether or not the individual variable values are positive or negative. The constraints for the other intermediate nodes are similarly constructed. We have no sign restrictions on the decision variables, rather we have the box constraints found earlier. The reader should verify that the appropriate LP for answering ExaByte's questions is:

$$\boxed{\begin{array}{ll} \max\limits_{x} & z = c^T x \\ s.t. & Ax = b \\ & \ell \leq x \leq u \\ & x \in \mathbb{R}^{10} \end{array}}$$

where

$$\boxed{\begin{array}{l}
x^T = \begin{bmatrix} x_{sa} & x_{sb} & x_{sc} & x_{ab} & x_{ad} & x_{be} & x_{cd} & x_{ce} & x_{dt} & x_{et} \end{bmatrix} \\[2ex]
c^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \\[2ex]
\ell^T = \begin{bmatrix} 0 & 0 & 0 & -12 & -10 & -28 & -41 & -35 & 0 & 0 \end{bmatrix} \\[2ex]
u^T = \begin{bmatrix} 32 & 13 & 9 & 12 & 10 & 28 & 41 & 35 & 43 & 8 \end{bmatrix} \\[2ex]
A = \begin{bmatrix} +1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & +1 & 0 & +1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & +1 & 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & +1 & 0 & +1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & +1 & 0 & +1 & 0 & -1 \end{bmatrix} \\[2ex]
b^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}
\end{array}}$$

The optimal solution is

$$x^* = \begin{bmatrix} 22 & 13 & 9 & 12 & 10 & 25 & 33 & -24 & 43 & 1 \end{bmatrix}^T$$

$$z^* = 44.$$

The network path is shown in Figure 4.4. The maximum transfer rate is 44. Notice that $x_{ce} < 0$ so data moves from node e to node c, while $x_{ad} > 0$ so data moves from node a to node d.

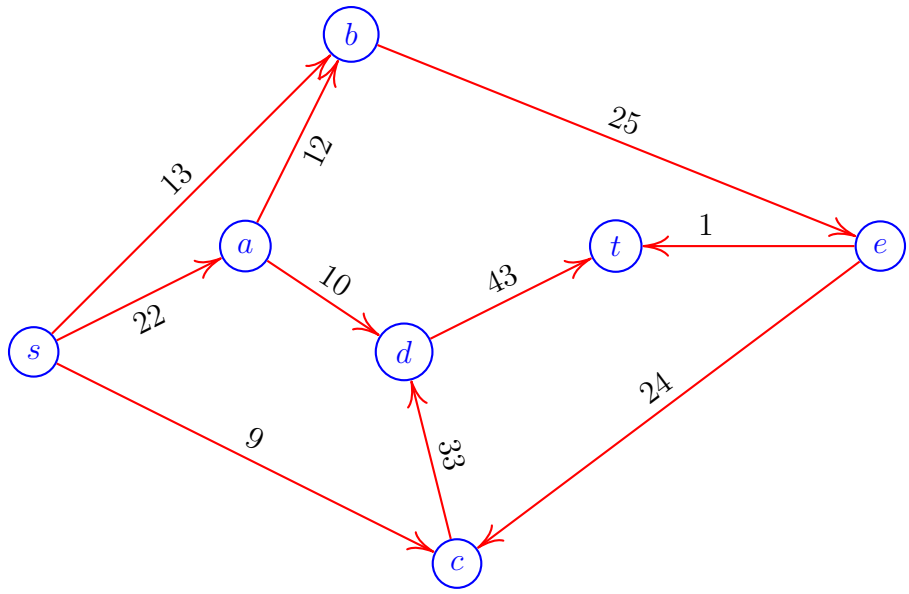


Figure 4.4: Solution of the ExaByte computer network problem with transfer rates along each connection indicated by arrows for direction and numerical values for transfer rates.

Remark 21. *If the bounds on every variable are captured using bound constraints $\ell \leq x \leq u$, then one should always use the lower and upper bound inputs provided by most software. Placing such constraints in the A matrix for inequality constraints $Ax \leq b$ is very inefficient for solvers and it is also an inefficient way to express these simple constraints.*

4.10 Counterfeit Bank Note Detection

Similarly to data fitting, linear programming can also be used to find an optimal nonlinear functions that separate data into two classes. This model objective is to maximize a quantity that can only be expressed in terms of inequalities.

MegaBank is studying methods for detection of counterfeit bank notes. Because of the volume of bills that pass through the bank daily, they wish to have an automated system based on analysis of visual clues. One strategy is to examine properties of the histogram of wavelet coefficients which describe an image of a banknote. Histograms of authentic bills and counterfeit bills show distinct properties.[2] Sample data is shown in Figure 4.5. Wavelet coefficient histogram analysis was performed for images of 75 authentic bank notes (blue) and 75 counterfeit bank notes (red). The two groups form distinct sets in the given two-dimensional space. MegaBank would like a simple function that separates the two groups so that if a new bill has histogram properties below this function, it will be flagged as a likely counterfeit.

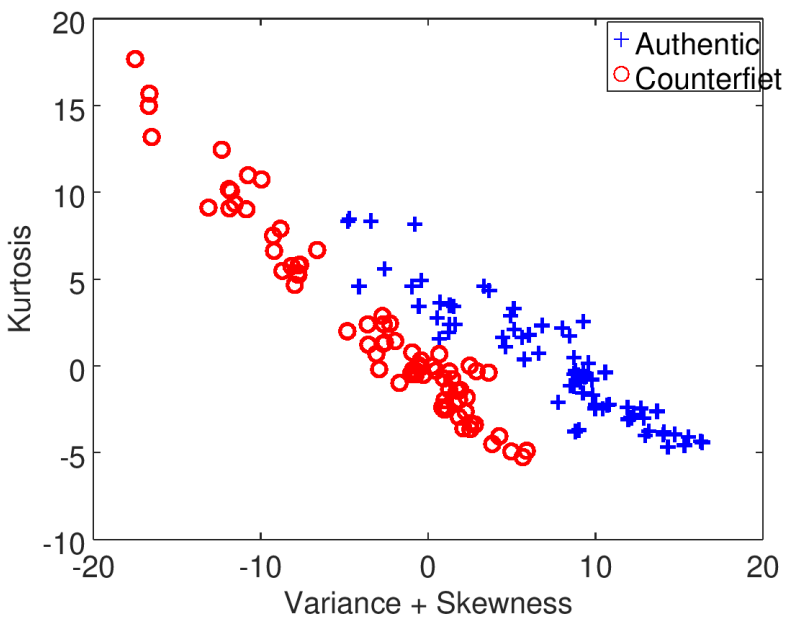


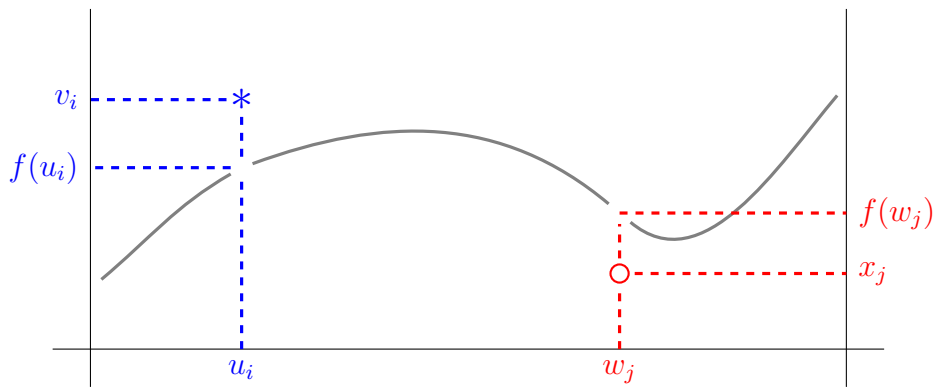
Figure 4.5: Wavelet coefficient histogram data for images of 75 authentic bank notes (blue +) and 75 counterfeit bank notes (red ∘).

Solution. Each bank note is associated with a point (u, v), where the variable u indicates the sum of the bank note's histogram variance and skewness, and the variable

[2] L. Volker, J. L. Hoffmann, H. Dorksen, R. Hildebrand, E. Gillich, J. Hofmann, J. Schaede, *Banknote authentication with mobile devices*, Proc.SPIE, 8665, (2013).

v indicates the bank note's histogram kurtosis. We seek a function $f(u)$ so that if the bank note is authentic $f(u) \leq v$, but if the bank note is counterfeit $f(u) > v$.

Let (u_i, v_i) be the set of points associated with m authentic bank notes. Let (w_j, x_j) bet the set of points associated with n counterfeit bank notes. Consider the following picture.



The '*' indicates a point (u_i, v_i) which should lie above the function f (solid line). The 'o' indicates a point (w_j, x_j) which should like below the function f. If all points (u_i, v_i) lie above the function f by at least some value δ, then

$$v_i - f(u_i) \geq \delta \qquad i = 1, 2, \ldots, m.$$

Similarly, if all points (w_j, x_j) lie below the function f by at least some value δ, then

$$f(w_j) - x_j \geq \delta \qquad j = 1, 2, \ldots, n.$$

One possible good separator function is the one that maximizes the gap δ. Consider functions that can be written as a linear combination of N given arbitrary functions ϕ_k:

$$f(x) = \sum_{k=1}^{N} a_k \phi_k(x),$$

where the a_k are coefficients to be determined. We have the linear program

$$\boxed{\begin{array}{lll} \max\limits_{\delta, a} & z = \delta & \\ s.t. & v_i - \displaystyle\sum_{k=1}^{N} a_k \phi_k(u_i) \geq \delta & i = 1, 2, \ldots, m \\ & \displaystyle\sum_{k=1}^{N} a_k \phi_k(w_j) - x_j \geq \delta & j = 1, 2, \ldots, n \\ & \delta \in \mathbb{R} & \\ & a \in \mathbb{R}^N. & \end{array}}$$

We can write this LP in matrix form:

$$\boxed{\begin{array}{ll} \max\limits_{y} & z = c^T y \\ s.t. & Ay \leq B \\ & y \in \mathbb{R}^{N+1} \\ & \\ & y^T = \begin{bmatrix} \delta & a_1 & a_2 & \dots & a_N \end{bmatrix} \\ & \\ & c^T = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \end{bmatrix} \\ & \\ & A = \begin{bmatrix} \mathbf{1_m} & \phi_1(u) & \phi_2(u) & \dots & \phi_N(u) \\ \mathbf{1_n} & -\phi_1(w) & -\phi_2(w) & \dots & -\phi_N(w) \end{bmatrix} \\ & \\ & B = \begin{bmatrix} v \\ -x \end{bmatrix} \end{array}}$$

Keep in mind that each $\phi_k(u_i)$ and each $\phi_k(w_j)$ are known values containing not decision variables (δ and a).

Linear Classifier. Suppose we wish to solve MegaBank's problem using a linear classifier: $f(t) = a_1 + a_2 t$. In this scenario, we have $N = 2$, $\phi_1(t) = 1$ and $\phi_2(t) = t$. We also have $n = m = 75$. The LP is:

$$\boxed{\begin{array}{ll} \max\limits_{y} & z = c^T y \\ s.t. & Ay \leq B \\ & y \in \mathbb{R}^{3} \\ & \\ & y^T = \begin{bmatrix} \delta & a_1 & a_2 \end{bmatrix} \\ & \\ & c^T = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \\ & \\ & A = \begin{bmatrix} \mathbf{1_n} & \mathbf{1_n} & u \\ \mathbf{1_n} & -\mathbf{1_n} & -w \end{bmatrix} \\ & \\ & B = \begin{bmatrix} v \\ -x \end{bmatrix} \end{array}}$$

The optimal solution is $\delta^* = -0.832$, $a_1^* = 1.87$ and $a_2^* = -0.86$. That is, the optimal classifier function is $f(t) = 1.87 - 0.86t$. This function, is shown in the top-left plot in Figure 4.6. Notice that this classifier function does not satisfy the condition that all points for authentic bank notes lie above and all points for counterfeit bank notes lie below. This is because no such linear classifier exists. The maximum gap δ^* is

negative. In this case, δ measures the largest mis-classification, and the best fit line is the one for which the mis-classification is minimal for both types of bank notes.

Higher-Order Classifiers. Optimal classifier functions are shown in Figure 4.6 for polynomial orders up to five. The quadratic classifier also does not separate all data points into their known respective groups ($\delta^* = -0.341$). However, the higher-order classifiers succeed in dividing the data into the two known groupings ($\delta^* > 0$).

The usefulness of any classifier is not solely determined by how well it can divide known data into distinct groups. A classifier should also accurately place new data points into their correct groups. We can measure the effectiveness of a classifier by the false positive and false negative rates for data that were not used to construct the classifier. Consider 400 new data points, 200 for authentic bills and 200 for counterfeit bills. Using the same five classifiers, we find the following false classification data.

	linear	quadratic	cubic	quartic	quintic
false positives	17	11	1	1	5
false negatives	4	6	4	3	8

The geometry of the quintic classifier is significantly affected by borderline data points and does not accurately represent the nature of the boundary between data classes. It is a less accurate classifier than the classifiers of order three or four. Figure 4.7 shows the quintic classifier with the 400 test data points for which it performs poorly.

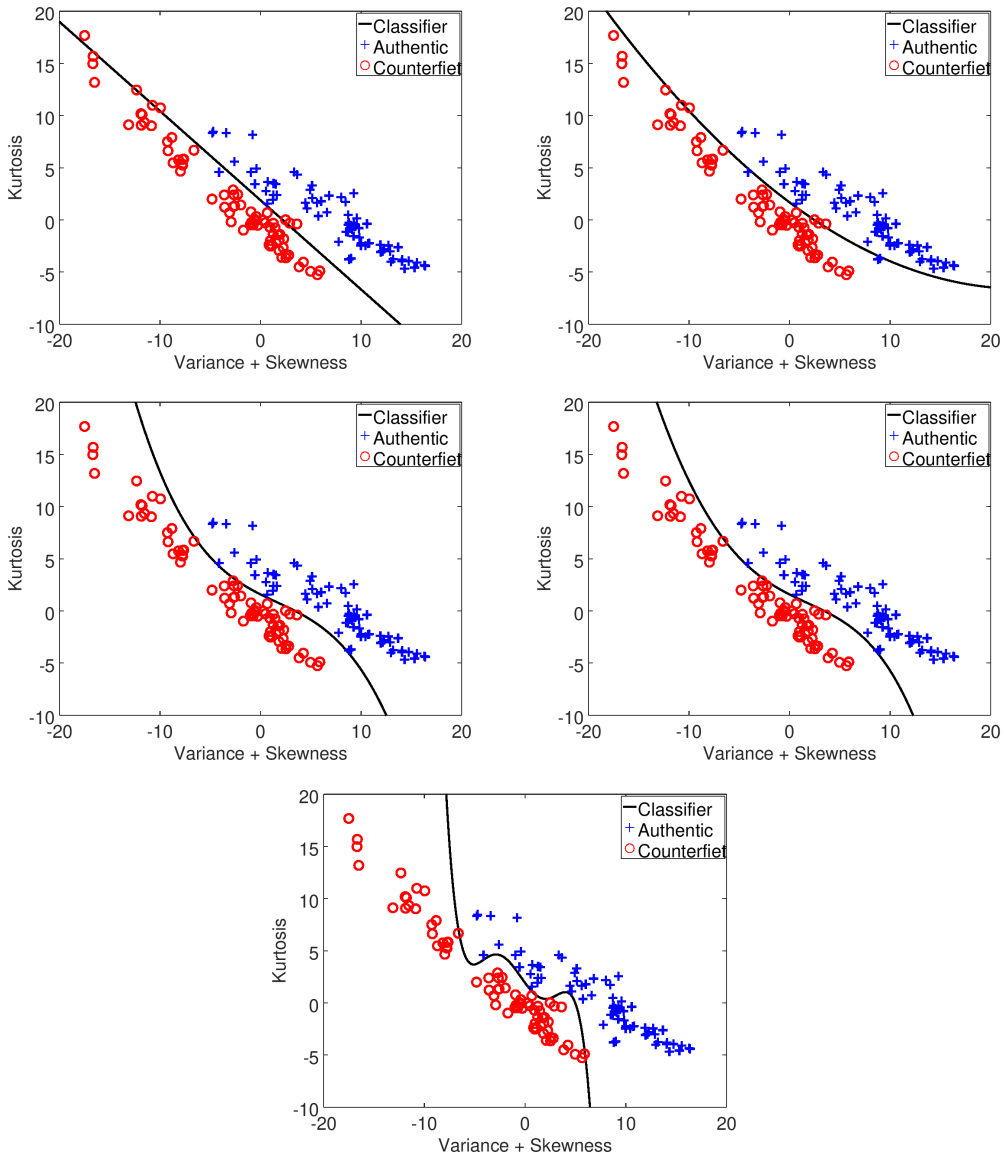


Figure 4.6: Five classifier functions for the MegaBank data set: linear (top-left), quadratic (top-right), cubic (middle-left), quartic (middle-right), quintic (bottom).

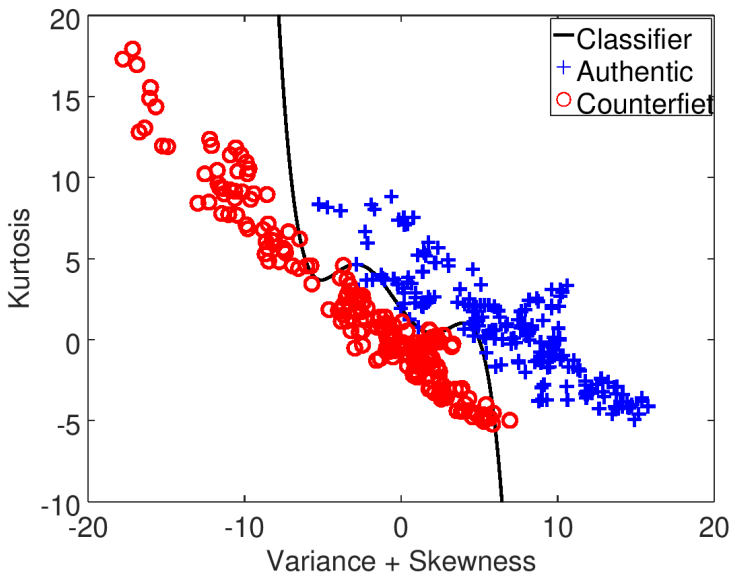


Figure 4.7: The quintic classifier shown with the set of 400 test data points.

4.11 Water Treatment Plant Planning

This example illustrates the formulation of fixed-fee or one-time costs that affect cost-base objectives. This is in contrast to per-item costs. We will also see our first use of binary variables that are only allowed to take on the specific values zero and one.

The state of Washington plans to build pollution removal facilities along the Columbia River. Four sites are under consideration. The cost of building at each site and the cost of treating each ton of water is given in the table. The legislature requires that at least 80,000 tons of pollutant 1 and 50,000 tons of pollutant 2 be removed from the river. The amounts of each pollutant removed from treated water are given in the following table. Find a lowest-cost strategy that meets the legislatures goals.

site	cost of building ($)	cost of treating 1 ton of water ($)	tons pollutant 1 per ton of treated water	tons pollutant 2 per ton of treated water
1	100,000	20	0.40	0.30
2	70,000	30	0.30	0.20
3	60,000	30	0.25	0.20
4	40,000	40	0.20	0.25

Solution. Notice that there is no requirement to build a treatment plant on any particular site. First, construct a linear or integer program that takes into account all of the problem conditions *except* for deciding where to build. That is, assume that the facilities are constructed free of cost. Let

$$x_i = (\text{tons of water processed at site } i), \qquad i = 1, 2, 3, 4.$$

We wish to minimize the cost of water treatment. We have the (initial) integer program

$$\boxed{\begin{array}{ll} \min\limits_{x} & z = 20x_1 + 30x_2 + 30x_3 + 40x_4 \\ s.t. & 0.40x_1 + 0.30x_2 + 0.25x_3 + 0.20x_4 \geq 80{,}000 \\ & 0.30x_1 + 0.20x_2 + 0.20x_3 + 0.25x_4 \geq 50{,}000 \\ & x \geq 0 \\ & x \in \mathbb{R}^4 \end{array}}$$

Next, we consider how to modify this initial LP to account for the cost of building facilities. Because we do not need to build on each site, we create new binary variables.

For $i = 1, 2, 3, 4$, we define

$$y_i = \begin{cases} 1 & \text{if a facility is built on site } i \\ 0 & \text{if a facility is not built on site } i \end{cases}$$

Our remaining task is to create appropriate constraints that connect the original variables x with the new variables y. The constraints need to model the condition:

$$y_i = \begin{cases} 1 & \text{if } x_i > 0 \\ 0 & \text{if } x_i = 0 \end{cases}$$

If we can do this, then the new objective function which measures the total cost is

$$z = 20x_1 + 30x_2 + 30x_3 + 40x_4 + 100{,}000y_1 + 70{,}000y_2 + 60{,}000y_3 + 40{,}000y_4.$$

Notice that with this objective function, the necessary constraint conditions can be written simply $y_i = 1$ if $x_i > 0$, $y_i \in \{0, 1\}$. This is because, if $x_i = 0$ then the objective will force $y_i = 0$ in order to minimize the cost. That is, the optimization problem will never choose to build on a site that does not process any water. The necessary constraints are

$$y_i \geq \frac{x_i}{M}, \quad y_i \in \{0, 1\},$$

where M is chosen so that $0 \leq x_i \leq M$ for all possible x_i. Notice that if $x_i > 0$, then $y_i > 0$ and because $y_i \in \{0, 1\}$, we have $y_i = 1$. A sufficiently large value for M can be chosen based on existing problem constraints. Notice that the first constraint shows that

$$x_1 \leq 80{,}000/0.40 = 200{,}000,$$

$$x_2 \leq 80{,}000/0.30 < 267{,}000,$$

$$x_3 \leq 80{,}000/0.25 = 320{,}000,$$

$$x_4 \leq 80{,}000/0.20 = 400{,}000.$$

Thus, $M = 400{,}000$ is sufficient for any combination of nonzero values for x. Similarly, the second constraint shows that

$$x_1 \leq 50{,}000/0.30 < 167{,}000,$$

$$x_2 \leq 50{,}000/0.20 = 250{,}000,$$

$$x_3 \leq 50{,}000/0.20 = 250{,}000,$$

$$x_4 \leq 50{,}000/0.25 = 200{,}000.$$

We see that $M = 250{,}000$ is sufficient for any combination of nonzero values for x. Overall, we can choose $M = 400{,}000$ as a good upper bound on x. Thus, the full integer program is

$$
\begin{aligned}
\min_{x,y} \quad & z = 20x_1 + 30x_2 + 30x_3 + 40x_4 + 100{,}000y_1 + 70{,}000y_2 + 60{,}000y_3 + 40{,}000y_4 \\
s.t. \quad & 0.40x_1 + 0.30x_2 + 0.25x_3 + 0.20x_4 \geq 80{,}000 \\
& 0.30x_1 + 0.20x_2 + 0.20x_3 + 0.25x_4 \geq 50{,}000 \\
& x_i - 400{,}000y_i \leq 0 \qquad i = 1, 2, 3, 4 \\
& x \geq 0 \\
& x \in \mathbb{R}^4 \\
& y_i \in \{0, 1\} \qquad i = 1, 2, 3, 4
\end{aligned}
$$

Remark 22. *In a cost minimization problem with fixed fee k for initiating some action $x > 0$, we introduce a binary variable y which indicates the presence (or absence) of the fixed cost. We have an objective function $z = f(x) + ky$ with constraint $x - My \leq 0$ for a value M larger than any possible value for x.*

Writing the binary constraints using linear inequalities, we have the equivalent mixed-integer program

$$
\begin{aligned}
\min_{x,y} \quad & z = 20x_1 + 30x_2 + 30x_3 + 40x_4 + 100{,}000y_1 + 70{,}000y_2 + 60{,}000y_3 + 40{,}000y_4 \\
s.t. \quad & 0.40x_1 + 0.30x_2 + 0.25x_3 + 0.20x_4 \geq 80{,}000 \\
& 0.30x_1 + 0.20x_2 + 0.20x_3 + 0.25x_4 \geq 50{,}000 \\
& x_i - 400{,}000y_i \leq 0 \qquad i = 1, 2, 3, 4 \\
& 0 \leq x \\
& 0 \leq y \leq 1 \\
& x \in \mathbb{R}^4 \\
& y \in \mathbb{Z}^4
\end{aligned}
$$

4.12 Goldbach's Conjecture

This example illustrates the selection a subset of items from a larger set. The subset must conform to specific criteria. Sometimes the selection must also optimize some merit criterion. We will also see our first example of a "feasibility problem" in which the objective function is not relevant to finding a solution.

In 1752, the mathematician Christian Goldbach made a conjecture concerning prime numbers. In modern terms, his, as yet unproven, conjecture is written: "Every even number greater than 2 can be written as the sum of two prime numbers." While we are not about to prove or disprove this conjecture, we can construct an integer program that can test the conjecture for a particular choice of even number. For example, given the even number 10, our integer program may return the two prime numbers 7 and 3 because $10 = 3 + 7$.

Suppose we seek two prime numbers which when added together sum to a given even number E. The candidate prime numbers must come from the ordered set of prime numbers $\{p_k\}_{k=1}^{N}$ where p_N is the largest prime less than E. For example, suppose $E = 34$, then the ordered set of prime numbers for this scenario is

$$\{p_k\}_{k=1}^{11} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31\},$$

and there are a few solutions: $34 = 3 + 31 = 5 + 29 = 11 + 23 = 17 + 17$.

The idea is to create an integer program that chooses which two prime numbers can sum to a given even number. Since our list of primes has N elements, we let x_k be the number of times p_k is included in the sum. Then $\sum_{k=1}^{N} x_k p_k$ is the sum of a selection of prime numbers and $\sum_{k=1}^{N} x_k$ is the number of primes selected. We have the initial integer optimization problem

$$\boxed{\begin{array}{ll} \min\limits_{x} & z = \left| E - \sum\limits_{k=1}^{N} x_k p_k \right| \\ s.t. & \sum\limits_{k=1}^{N} x_k = 2 \\ & x \geq 0 \\ & x \in \mathbb{Z}^N \end{array}}$$

If the prime sum is equal to E then we have a solution with $z^* = 0$. However, if no prime sum adds to E, then we have found an even integer E that does not satisfy Goldbach's conjecture. We can reformulate this optimization problem as an integer

program.

$$\boxed{\begin{array}{ll} \min\limits_{\delta,x} & z = \delta \\ s.t. & \sum\limits_{k=1}^{N} x_k = 2 \\ & \delta \geq +E - \sum\limits_{k=1}^{N} x_k p_k \\ & \delta \geq -E + \sum\limits_{k=1}^{N} x_k p_k \\ & x \geq 0 \\ & \delta \in \mathbb{Z} \\ & x \in \mathbb{Z}^N \end{array}}$$

For this problem, we seek a specific condition on the objective value – that it equals zero. So, we can alternatively reformulate this problem as a feasibility problem in which we seek *any* feasible point. To do this, we set $\delta = 0$, and our inequality constraints become an equality constraint:

$$\boxed{\begin{array}{ll} \min\limits_{x} & z = 0 \\ s.t. & \sum\limits_{k=1}^{N} x_k = 2 \\ & \sum\limits_{k=1}^{N} x_k p_k = E \\ & x \geq 0 \\ & x \in \mathbb{Z}^N \end{array}}$$

The objective function is no longer relevant because we already know what objective value we require. The only question is whether or not some set of decision variables is feasible with this value.

The following Octave code tests an even number E to see if there are two prime numbers with E as the sum. The user only selects the value of E.

```
E=62;            % choose a positive even integer
P=primes(E-1);   % P is the set of prime numbers (row vector)
N=length(P);     % N is the number of primes in P
c=zeros(N,1);
A=[ones(1,N);P];
b=[2;E];
lb=zeros(N,1);
ub=2*ones(N,1);
ct='SS';
vt=repmat('I',1,N);
sense=1;
[xstar,zstar,er,ex]=glpk(c,A,b,lb,ub,ct,vt,sense);

% These remaining lines produce nice output to the prompt

if ex.status==5
  idx=find(xstar);
  fprintf('%d = %d + %d\n',E,P(idx(1)),P(idx(end)));
else
  fprintf('No two primes sum to %d\n',E);
end
```

The corresponding Python code is as follows (with basic output).

```
import numpy as np
from sympy import primerange
E=62
P=list(primerange(2,E))
N=len(P)
P=np.array(P).reshape((1,N))
c=np.zeros((N,1))
Ae=np.concatenate((np.ones((1,N)),P),axis=0)
be=[[2],[E]]
bds=(((0,2), )*N)
res=opt.linprog(c,A_eq=Ae,b_eq=be,bounds=bds)
print(res.x)
```

4.13 Packing Books

This example demonstrates a variety of uses for binary variables involved in subset selection. With binary variables, simple constraints can encode high-level conditions such as if-then and either-or statements. A so-called greedy algorithm is also presented in contrast to the formulated IP. This algorithm illustrates how simple selection rules can fail to produce and optimal result, even with small problems.

Two friends are planning a cross country drive and are selecting which books to bring to read during the drive. The choices of books and their length are shown in the table below. They have decided to choose books based on the following restrictions.

- Bring no more than two of the three mystery stories.
- Bring at least one of the three books from the "... of the World" series.
- Bring at least two of the four books about cats.
- Either bring all three Kangaroo books or bring none.
- If "Short Stories for Long Trips" is brought then "Long Stories for Short Trips" should also be brought.
- "Math is Fun" must be brought if either "Shipwrecks of the World" or "The Santa Fe Mystery" is brought.
- Bring books whose lengths total as many pages as possible, but limit this length to no more than 3000 pages.

Formulate an integer program that can be used to select the books to take on the trip.

Book Number	Book Length (pages)	Title
1	231	The Mystery of the Golden Cat
2	156	Jokes of the world
3	462	A Climber's Guide to Volcanoes
4	312	The Kangaroo Cart Mystery
5	124	Kangaroo Breeds for Dummies
6	296	Math is Fun
7	446	The Santa Fe Mystery
8	374	Short Stories for Long Trips
9	228	Cats of the World
10	374	Long Stories for Short Trips
11	155	Shipwrecks of the World
12	362	Cats are Fun
13	397	Kangaroos and Cats, Friends for Life

Solution. We have 13 choices to make – whether or not to bring each of 13 books. For $k = 1, 2, \ldots, 13$, let

$$x_k = \begin{cases} 1 & \text{if book number } k \text{ is to be brought on the trip} \\ 0 & \text{if book number } k \text{ is not to be brought on the trip} \end{cases}$$

Also, let p_k be the number of pages in book k. Our goal is to maximize the total number of pages in books brought on the trip:

$$z = \sum_{k=1}^{13} p_k x_k,$$

subject to many constraints. Because all of the decision variables are binary, a variety of conditional constraints can be modeled simply. Let's consider the list in order.

- *Bring no more than two of the three mystery stories.*

  The three mystery stories are books 1, 4 and 7. So we limit these choices to any two or fewer:

  $$x_1 + x_4 + x_7 \leq 2.$$

  This constraint limits our mystery book choices to any combination of the three except all three together.

- *Bring at least one of the three books from the "... of the World" series.*

  Similarly to the mystery book constraint, we have

$$x_2 + x_9 + x_{11} \geq 1$$

- *Bring at least two of the four books about cats.*

$$x_1 + x_9 + x_{12} + x_{13} \geq 2.$$

- *Either bring all three Kangaroo books or bring none.*

  One method for ensuring that the three Kangaroo-related variables are all the same is to impose the simple set of constraints

$$x_4 = x_5 = x_{13}.$$

  Another method (which lends itself to interesting generalizations) is to introduce a new binary variable:

$$y = \begin{cases} 1 & \text{if all three Kangaroo books are brought on the trip} \\ 0 & \text{if no Kangaroo books are brought on the trip} \end{cases}$$

  Then, we have the constraint

$$x_4 + x_5 + x_{13} = 3y.$$

  Notice that when $y = 0$ it must be true that $x_4 = x_5 = x_{13} = 0$. And when $y = 1$ it must be true that $x_4 = x_5 = x_{13} = 1$. No other combinations of y, x_4, x_5, x_{13} satisfy this constraint.

- *If "Short Stories for Long Trips" is brought then "Long Stories for Short Trips" should also be brought.*

  This condition reads, if $x_8 = 1$ then $x_{10} = 1$, but if $x_8 = 0$ then x_{10} can equal either 0 or 1. This condition is met using the following constraint:

$$x_{10} \geq x_8.$$

  Remember: the x_k are restricted to be binary.

- *"Math is Fun" must be brought if either "Shipwrecks of the World" or "The Santa Fe Mystery" is brought.*

  Similarly to the previous constraint:

$$2x_6 \geq x_7 + x_{11}.$$

  If $x_7 + x_{11}$ has a value of 1 or 2, then $x_6 = 1$. Otherwise x_6 could equal either zero or one.

- *Bring books whose lengths total as many pages as possible, but limit this length to no more than 3000 pages.*

$$\sum_{k=1}^{13} p_k x_k \leq 3000.$$

Bringing the objective and constraints together, we have the binary integer program

$$\boxed{\begin{array}{ll} \min\limits_{x,y} & z = \sum\limits_{k=1}^{13} p_k x_k \\ s.t. & x_1 + x_4 + x_7 \leq 2 \\ & x_2 + x_9 + x_{11} \geq 1 \\ & x_1 + x_9 + x_{12} + x_{13} \geq 2 \\ & x_4 + x_5 + x_{13} - 3y = 0 \\ & x_8 - x_{10} \leq 0 \\ & 2x_6 - x_7 - x_{11} \geq 0 \\ & z = \sum\limits_{k=1}^{13} p_k x_k \leq 3000 \\ & 0 \leq y, x_k \leq 1 \qquad 1 \leq k \leq 13 \\ & y, x_k \in \mathbb{Z} \qquad 1 \leq k \leq 13 \end{array}}$$

The answer is to bring every book except books 1, 3 and 9 ("The Mystery of the Golden Cat," "A Climbers Guide to Volcanoes" and "Cats of the World"). The total page count is $z^* = 2996$.

A Greedy Algorithm Approach. Suppose we compare the optimal solution to a book choice determined by a greedy approach. The idea is the following.

1. Begin with the books on the shelf and an empty box of books to take on the trip.

2. Choose the book on the shelf with the largest number of pages.

3. Place this book in the box if its inclusion would not violate any of the constraints. Otherwise, set it aside.

4. If there is at least one book still on the shelf, then go to step 2. Otherwise, go to step 5.

5. If the box of books satisfies all constraints, then the box contains a greedy selection. Otherwise, the greedy algorithm failed to provide a feasible solution.

The following table shows the results of each pass of the greedy algorithm at step 3. Numerical values in the table are book numbers.

books on shelf	books in box (pages)	set aside	notes
1,2,3,4,5,6,7,8,9,10,11,12,13			Starting Condition
1,2,4,5,6,7,8,9,10,11,12,13	3 (462)		Longest book added to box
1,2,4,5,6,8,9,10,11,12,13	3,7 (908)		Next longest added
1,2,4,5,6,8,9,10,11,12	3,7,13 (1305)		Third longest added
1,2,4,5,6,9,10,11,12	3,7,8,13 (1679)		8 and 10 are same length
1,2,4,5,6,9,11,12	3,7,8,10,13 (2053)		8 and 10 are same length
1,2,4,5,6,9,11	3,7,8,10,12,13 (2415)		Next longest
1,2,5,6,9,11	3,4,7,8,10,12,13 (2722)		Next longest
1,2,5,9,11	3,4,7,8,10,12,13 (2722)	6	page constraint
2,5,9,11	3,4,7,8,10,12,13 (2722)	1,6	first constraint
2,5,11	3,4,7,8,9,10,12,13 (2950)	1,6	9 added to box
5,11	3,4,7,8,9,10,12,13 (2950)	1,2,6	page constraint
5	3,4,7,8,9,10,12,13 (2950)	1,2,6,11	page constraint
	3,4,7,8,9,10,12,13 (2950)	1,2,5,6,11	page constraint

Unfortunately for this algorithm, not only did it terminate with fewer total pages than the optimal solution, the books in the box do not satisfy the fourth and sixth constraints. In general, greedy algorithms cannot guarantee an optimal solution. Though, in practice, they can provide very good solutions using simple rules. Unfortunately, they can also result in very poor solutions – far from feasible and/or optimal.

4.14 CopyMaster Machine Scheduling

This example demonstrates how to linearize an objective function which is defined as the maximum on a set of given functions. The concept is to minimize the worst condition among many, rather than minimizing the sum. An absolute value objective is a special case of this result. This example also illustrates the use of binary variables in enforcing either/or box constraints.

CopyMaster is a company that specializes in making lots of copies. The company has m copy machines of varying capabilities and speed. The time required to complete any of n types of jobs varies with machine choice. Let c_{ij} be the machine time required to complete a job of type $j = 1, 2, \ldots, n$ using machine $i = 1, 2, \ldots, m$. Suppose CopyMaster has N_j jobs of each type j to complete. The machines can run concurrently. Can we use linear programming to decide how to assign jobs to machines in order to complete the jobs as quickly as possible? This week begins with the machine data and job requirements listed in the following tables. Times c_{ij} are in minutes.

c_{ij}	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	$j=6$
$i=1$	6	12	8	12	10	3
$i=2$	8	6	12	7	6	10
$i=3$	8	11	6	6	7	4
$i=4$	3	9	8	10	3	10

	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	$j=6$
N_j	641	190	386	469	395	482

Solution. Our decision is to determine how many of each job to assign to each machine. We will assume that breaking up jobs among machines does not significantly influence the total work time. Let

$$x_{ij} = (\text{number of jobs of type } j \text{ run on machine } i)$$

$$x_{ij} \in \mathbb{Z}, \quad x_{ij} \geq 0$$

$$i = 1, 2, 3, 4 \quad \text{and} \quad j = 1, 2, 3, 4, 5, 6.$$

The objective function must measure the time it takes the busiest machine to complete its set of jobs. The time that machine i is busy with its jobs is

$$\sum_{j=1}^{6} c_{ij} x_{ij}.$$

So, the time of the busiest machine is

$$z = \max_{i=1,2,3,4} \left(\sum_{j=1}^{6} c_{ij} x_{ij} \right) = \max \left\{ \sum_{j=1}^{6} c_{1j} x_{1j} \ , \ \sum_{j=1}^{6} c_{2j} x_{2j} \ , \ \sum_{j=1}^{6} c_{3j} x_{3j} \ , \ \sum_{j=1}^{6} c_{4j} x_{4j} \right\}.$$

The objective is to *minimize* this function. If we introduce a new variable T as the time when the busiest machine completes its jobs, we have the equivalent linear formulation

$$\min_{x,T} z = T \ \text{ s.t. } \sum_{j=1}^{6} c_{ij} x_{ij} \leq T, \ i = 1, 2, 3, 4.$$

We also have the process constraints that make sure that all jobs are completed.

$$\sum_{i=1}^{4} x_{ij} = N_j, \quad j = 1, 2, 3, 4, 5, 6.$$

We have the complete integer program

$$\boxed{\begin{array}{ll} \min\limits_{x,T} & z = T \\ s.t. & \sum\limits_{j=1}^{6} c_{ij} x_{ij} - T \leq 0, \qquad i = 1, 2, 3, 4 \\ & \sum_{i=1}^{4} x_{ij} = N_j \qquad j = 1, 2, 3, 4, 5, 6 \\ & x_{ij}, T \geq 0 \\ & x_{ij}, T \in \mathbb{Z} \end{array}}$$

The optimal solution is for the jobs to be divided among the machines according to the following table.

x_{ij}^*	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$
$i = 1$	72		126			482
$i = 2$		190		248	2	
$i = 3$			260	221		
$i = 4$	569				393	

Machines 1, 3 and 4 run a total of 2886 minutes each. Machine 2 runs for $z^* = T^* = 2888$ minutes (48 hours and 8 minutes).

Octave Code. This optimization problem has constraints that contain a lot of structure. A relatively short piece of Octave code can construct large matrices used by the LP solver. The user need only change the two inputs C and N.

```
% Enter the Cij matrix data (this could be automated)
C=[6 12  8 12 10  3
8  6 12  7  6 10
8 11  6  6  7  4
3  9  8 10  3 10];

% Enter the number of jobs data
N=[641;190;386;469;395;482];

% Create some size variables
[NumMachines,NumJobs]=size(C);
n=NumMachines*NumJobs+1;
M=max(C(:))*max(N);

% Create the objective vector
c=[zeros(n-1,1);1];

% Create the A matrix
A1=kron(ones(1,NumJobs),eye(NumMachines));
idx=find(A1);
A1(idx)=C;
A1=[A1 -ones(NumMachines,1)];
A2=kron(eye(NumJobs),ones(1,NumMachines));
A2=[A2 zeros(NumJobs,1)];
A=[A1;A2];

% Create the rhs b vector
b=[zeros(NumMachines,1);N];

% Create the various problem variables for the solver
lb=zeros(n,1);
ub=M*ones(n,1);
ct=[repmat('U',1,NumMachines) repmat('S',1,NumJobs)];
vt=repmat('I',1,n);
sense=1;

% Call the solver
[x,z]=glpk(c,A,b,lb,ub,ct,vt,sense);

% Report the solution
reshape(x(1:end-1),NumMachines,NumJobs)
```

Either/Or Box Constraints. One potential difficulty with the optimal solution for the CopyMaster problem is that very small subjobs can occur. For example, the set of jobs of type $j = 5$ is divided into subjobs of 393 and 2. In order to improve the logistics of dividing and re-combining job types, suppose constraints are added to enforce a lower bound on the size of subjobs at 24. That is, either $x_{ij} = 0$ or $x_{ij} \geq 24$. If we introduce new binary variables

$$y_{ij} = \begin{cases} 1 & \text{if machine } i \text{ processes some of job } j \\ 0 & \text{if not} \end{cases}$$

then we can formulate the needed constraints as

$$24y_{ij} \leq x_{ij} \leq My_{ij}.$$

Notice that when $y_{ij} = 0$ we have $x = 0$ and when $y_{ij} = 1$ we have $24 \leq x_{ij} \leq M$. The only remaining task is to choose M to be larger than any feasible x_{ij}. In this case, M can equal the maximum number of jobs of any type, $M = 641$.

This type of constraint can be generalized to limit any decision variable to be within one of several intervals. See Exercise 18.

4.15 Profit Under Premium Fees

This example illustrates how to reformulate continuous piecewise linear objective functions. The method is to recast the problem into the form of minimization of an objective defined as the maximum over functions. We also see careful use of changing problems from minimization to maximization to suit problem needs.

Sumi is planning a flight to Seattle where she will sell some of her remaining art pieces. She has the option to take as many pieces for which she can pay the shipping cost. The weight and selling price for each type of item are shown in the table below. Shipping volume is not a limiting consideration, but shipping weight is a concern. Shipping costs are as follows: $0 - 50$ pounds, \$2/lb; $50 - 100$ pounds, \$100+\$4/lb for every pound above 50; 100+ pounds, \$300+\$10/lb for every pound above 100 pounds. What is the maximum profit Sumi can realized on this venture and how many of each item should she take to Seattle?

index	item	shipping weight(lb)	selling price (\$)
1	oil painting	13	55
2	wood sculpture	40	150
3	batik wall hanging	18	75
4	glazed ceramic vase	25	100

Solution. The objective is to maximize profit. The simplest decision variables are to let

$$x_k = \text{(number of items of type } k \text{ to pack)}$$

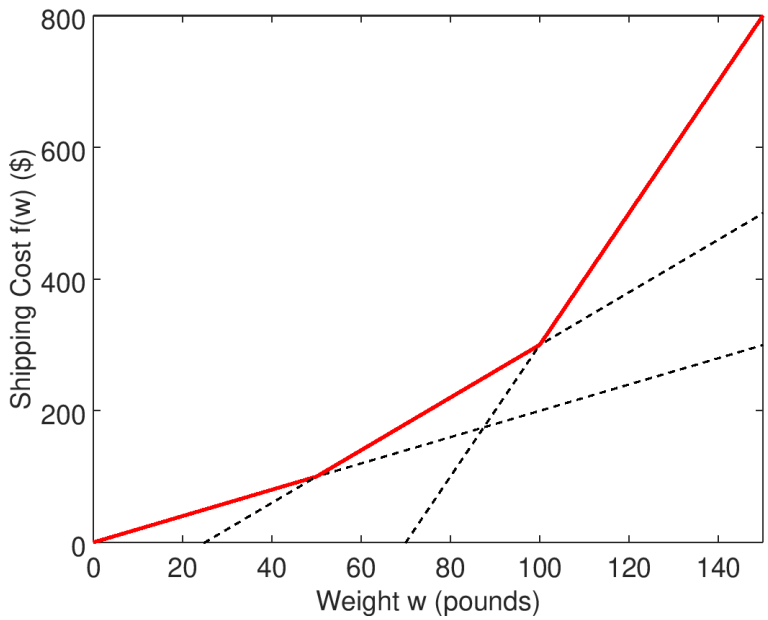
Then the profit function is

$$\begin{aligned} z &= \text{(profit \$)} \\ &= \text{(selling income \$) - (shipping cost \$)} \\ &= 55x_1 + 150x_2 + 75x_3 + 100x_4 - f(w), \end{aligned}$$

where the shipping cost $f(w)$ is a nonlinear function of weight $w = 13x_1 + 40x_2 + 18x_3 + 25x_4$. We have

$$f(w) = \begin{cases} 2w & \text{if } 0 \le w \le 50 \\ 100 + 4(w - 50) & \text{if } 50 < w \le 100 \\ 300 + 10(w - 100) & \text{if } 100 < w \end{cases}$$

A graph of this function shows that it is piecewise linear.



What this graph also shows is that the cost for any weight w is the minimum value that lies at or above *each* of the three lines. So, the shipping cost for any weight is the solution to

$$\begin{aligned} \min_{\delta} \quad & z = \delta \\ \text{s.t.} \quad & \delta \geq 2w \\ & \delta \geq 100 + 4(w - 50) \\ & \delta \geq 300 + 10(w - 100), \end{aligned}$$

or equivalently

$$\begin{aligned} \max_{\delta} \quad & z = -\delta \\ \text{s.t.} \quad & \delta \geq 2w \\ & \delta \geq 100 + 4(w - 50) \\ & \delta \geq 300 + 10(w - 100), \end{aligned}$$

That is, the shipping cost for any weight is δ^*. So, we can use this solution to the nonlinear shipping cost to modify the objective of Sumi's problem. We arrive at the

integer program:

$$\boxed{\begin{array}{ll} \max\limits_{x,w,\delta} & z = 55x_1 + 150x_2 + 75x_3 + 100x_4 - \delta \\ s.t. & w = 13x_1 + 40x_2 + 18x_3 + 25x_4 \\ & \delta \geq 2w \\ & \delta \geq 100 + 4(w - 50) \\ & \delta \geq 300 + 10(w - 100) \\ & x, w, \delta \geq 0 \\ & x \in \mathbb{Z}^4 \\ & w, \delta \in \mathbb{Z} \end{array}}$$

The optimal solution is

$$x_1^* = 6, x_2^* = 0, x_3^* = 1, x_4^* = 0, w^* = 96, \delta^* = 284, z^* = 121.$$

Sumi should take six oil paintings and one batik wall hanging. The shipping weight of the art items will be 96 pounds and the shipping cost will be $\delta^* = \$284$. Sumi's profit will be \$121.

4.16 AJ Bread Distributors

This example show the linear programming formulation of a bipartite graph transportation problem. It illustrates how to balance a supply and demand problem through the use of virtual sources or virtual destinations. We also exercise our tools in fixed-fees, binary variables and summation notation. The problems are formulated entirely with the use of parameters rather than specific numerical values.

The AJ Bread Company has m bakeries which have produced $s_1, s_2, \ldots s_m$ batches of bread loaves. AJ Bread is contracted to deliver batches of bread in quantities $d_1, d_2, \ldots, d_n$ to n retail stores. Suppose (for now) that any number of batches is easily shipped from any bakery to any store. Transporting one batch of bread from bakery i to store j has a known cost c_{ij}. The c_{ij} need not be identical values due to, for example, higher costs associated with longer travel times or other unequal logistical conditions. What is the lowest cost strategy for supplying each store with its daily bread?

This problem is an example of a transportation problem. There are other efficient methods for solving these types of problems that do not use linear programming.

Solution. Since each bakery can supply bread to any store, we define

$$x_{ij} = (\text{number of batches shipped from bakery } i \text{ to store } j),$$

where $x_{ij} \geq 0$ and $x_{ij} \in \mathbb{Z}$. Each possible transportation path has an associated cost c_{ij}. Notice that

$$\sum_{j=1}^{n} x_{ij} = (\text{total batches originating from bakery } i),$$

and

$$\sum_{i=1}^{m} x_{ij} = (\text{total batches arriving at store } j).$$

Recognizing that we must meet consumer demands at each store and not exceed the baking capacity of any bakery, we have the compactly expressed integer program to

solve the AJ Bread Company problem:

$$\boxed{\begin{array}{lll} \min\limits_{x} & z = \sum\limits_{i=1}^{m}\sum\limits_{j=1}^{n} c_{ij}x_{ij} & \\ s.t. & \sum\limits_{j=1}^{n} x_{ij} \leq s_i & i = 1, 2, \ldots, m \\ & \sum\limits_{i=1}^{m} x_{ij} \geq d_j & j = 1, 2, \ldots, n \\ & x \geq 0 & \\ & x \in \mathbb{Z}^{nm} & \end{array}}$$

Balancing the AJ Bread Distribution. One interesting aspect of the distribution problem is that there is no need to ship more batches than are required to meet consumer demand. Let $s = \sum_{i=1}^{m} s_i$ be the total number of batches of bread at the bakeries, and let $d = \sum_{j=1}^{n} d_j$ be the total consumer demand. Consider the following three possibilities:

1. If $s < d$ then the available bread is insufficient to meet consumer demand. Thus, the IP is infeasible – no routing of bread will meet consumer demand.

2. If $s = d$ then we have what is referred to as a "balanced" problem. The bread supply is exactly sufficient to meet consumer demand. And here is the important observation: All of the supply and demand constraints will be active at any feasible solution. In this case we have the equivalent (standard form) problem:

$$\boxed{\begin{array}{lll} \min\limits_{x} & z = \sum\limits_{i=1}^{m}\sum\limits_{j=1}^{n} c_{ij}x_{ij} & \\ s.t. & \sum\limits_{j=1}^{n} x_{ij} = s_i & i = 1, 2, \ldots, m \\ & \sum\limits_{i=1}^{m} x_{ij} = d_j & j = 1, 2, \ldots, n \\ & x \geq 0 & \\ & x \in \mathbb{Z}^{nm} & \end{array}}$$

3. If $s > d$ then the bread supply exceeds the current demand. In this case, we seek to formulate a balanced problem *by cleverly introducing slack variables.* Create a virtual store, numbered $n + 1$, with demand exactly equal to the number

of excess batches, $d_{n+1} = s - d$. Then allow decision variables which allow "shipping" to the virtual destination. We have the balanced problem

$$\boxed{\begin{array}{lll} \min\limits_{x} & z = \sum\limits_{i=1}^{m}\sum\limits_{j=1}^{n+1} c_{ij}x_{ij} & \\ \text{s.t.} & \sum\limits_{j=1}^{n+1} x_{ij} = s_i & i = 1, 2, \ldots, m \\ & \sum\limits_{i=1}^{m} x_{ij} = d_j & j = 1, 2, \ldots, n+1 \\ & x \geq 0 & \\ & x \in \mathbb{Z}^{(n+1)m} & \\ \text{with} & \sum\limits_{i=1}^{m} s_i - \sum\limits_{j=1}^{n} d_j = d_{n+1}. & \end{array}}$$

In this scenario, we have new variables $x_{i,n+1}$ and new constants $c_{i,n+1}$. One interpretation of these quantities stems from the fact that we do not actually ship to the virtual destination. $x_{i,n+1}$ can be the number of batches of bread left at bakery i, and $c_{i,n+1}$ can be the cost of storing a batch of bread at bakery i.

Fixed-Cost Transportation. Suppose that there are fixed costs for transporting bread. For example, if AJ Bread chooses to send some bread from bakery i to store j, there may be a truck rental fee regardless of how many batches are sent. Let e_{ij} be such a fixed fee. In the familiar way, we define

$$y_{ij} = \begin{cases} 1 & \text{if } x_{ij} > 0 \\ 0 & \text{if } x_{ij} = 0 \end{cases}$$

Then, for sufficiently large value of M, say $M = \max_i\{s_i\}$, we have the integer program

$$\boxed{\begin{array}{ll}
\min\limits_{x,y} & z = \sum\limits_{i=1}^{m}\sum\limits_{j=1}^{n+1}(c_{ij}x_{ij} + e_{ij}y_{ij}) \\
s.t. & \sum\limits_{j=1}^{n+1} x_{ij} = s_i \qquad i = 1,2,\ldots,m \\
 & \sum\limits_{i=1}^{m} x_{ij} = d_j \qquad j = 1,2,\ldots,n+1 \\
 & x_{ij} - My_{ij} \leq 0 \qquad i = 1,2,\ldots,m;\ j = 1,2,\ldots,n+1 \\
 & x \geq 0 \\
 & x \in \mathbb{Z}^{(n+1)m} \\
 & y_{ij} \in \{0,1\}^{(n+1)m} \\
\text{with} & \sum\limits_{i=1}^{m} s_i - \sum\limits_{j=1}^{n} d_j = d_{n+1} \\
 & M = \max_i\{s_i\}
\end{array}}$$

The various quantities in this integer program are interpreted as follows (for $i = 1,2,\ldots,m$ and $j = 1,2,\ldots,n$).

- s_i: The number of batches of bread at bakery i that are available for shipping.
- d_j: The number of batches of bread which are in demand at store j.
- x_{ij}: The number of batches of bread transported from bakery i to store j.
- $x_{i,n+1}$: The number of batches of bread kept at bakery i.
- $c_{i,j}$: The per batch cost of transporting bread from bakery i to store j.
- $c_{i,n+1}$: The per batch cost of keeping bread at bakery i.
- e_{ij}: The fixed cost of transporting bread from bakery i to store j.
- $e_{i,n+1}$: The fixed cost of keeping bread at bakery i.

4.17 Puzzles and Games Inc.

This example illustrated how to treat a single pair if general either/or inequality constraints. The method incorporates an introduced binary variable and a corresponding sufficiently large value. Generalizations of the method are left as exercises.

Puzzles and Games Inc. has the facilities for manufacturing a variety of small toys and games. However, their operation is small enough in scale that during any one production period they need to choose subsets of products which can be manufactured. For example, suppose the factory can make some combination of puzzles and yoyos, OR some combination of boomerangs and frisbees. In the table below are the items, the expected profit per item, the units of raw materials need to make each toy or game, and the amount of raw materials in stock. How can Puzzles and Games Inc. use its stock materials to maximize profit during the current production cycle?

item	item #	profit	wood	glue	paint	plastic
puzzle	1	5	6	1	2	0
yoyo	2	4	5	0	3	3
boomerang	3	7	9	1	3	2
frisbee	4	8	0	0	3	9
(stock)			102	12	25	57

Solution. One strategy is to solve two optimization problems: one in which only puzzles and yoyos are manufactured, and another one in which only boomerangs and frisbees are manufactured. The optimal solution is then the one of higher profit. However, we wish to find a more general procedure that can be used to treat multiple such constraint simultaneously. First we can formulate the problem as if any combination of the four products can be manufactured. This IP is

$$\boxed{\begin{array}{ll} \max\limits_{x} & z = 5x_1 + 4x_2 + 7x_3 + 8x_4 \\ \text{s.t.} & 6x_1 + 5x_2 + 9x_3 \leq 102 \\ & x_1 + x_3 \leq 12 \\ & 2x_1 + 3x_2 + 3x_3 + 3x_4 \leq 25 \\ & 3x_2 + 2x_3 + 9x_4 \leq 57 \\ & x \geq 0 \\ & x \in \mathbb{Z}^4 \end{array}}$$

where x_k is the number of toys of type k to manufacture. The remaining production constraint is that

$$\text{either } x_1 + x_2 = 0 \quad \text{or} \quad x_3 + x_4 = 0.$$

Now, because $x \geq 0$ for any feasible solution, we can rephrase this condition as

$$\text{either } x_1 + x_2 \leq 0 \quad \text{or} \quad x_3 + x_4 \leq 0.$$

Suppose that we need to enforce the general condition "either $f(x) \leq 0$ or $g(x) \leq 0$." We introduce a binary variable:

$$y = \begin{cases} 1 & \text{if } f(x) \leq 0 \\ 0 & \text{if } g(x) \leq 0 \end{cases}$$

Then the necessary constraints are (for sufficiently large M):

$$f(x) \leq M(1-y), \quad g(x) \leq My, \quad y \in \{0,1\}.$$

Notice that if $y = 1$ we have the constraints

$$f(x) \leq 0, \quad g(x) \leq M,$$

and if $y = 0$ we have the constraints

$$f(x) \leq M, \quad g(x) \leq 0.$$

The key is to choose M large enough that the constraint $g(x) \leq M$ is never active when $y = 1$ and so that $f(x) \leq M$ is never active when $y = 0$.

For the Puzzles and Games problem we have $f(x) = x_1 + x_2$ and $g(x) = x_3 + x_4$ so that the IP that solves the problem is

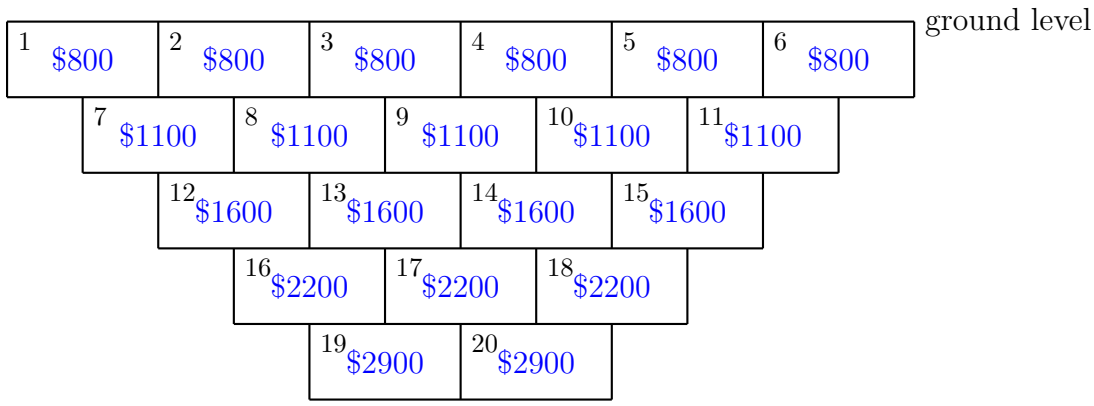
$$\boxed{\begin{array}{ll} \max\limits_{x,y} & z = 5x_1 + 4x_2 + 7x_3 + 8x_4 \\ \text{s.t.} & 6x_1 + 5x_2 + 9x_3 \leq 150 \\ & x_1 + x_3 \leq 12 \\ & 2x_1 + 3x_2 + 3x_3 + 3x_4 \leq 25 \\ & 3x_2 + 2x_3 + 9x_4 \leq 57 \\ & x_1 + x_2 \leq M(1-y) \\ & x_3 + x_4 \leq My \\ & x \geq 0 \\ & x \in \mathbb{Z}^4 \\ & y \in \{0,1\} \\ \text{with} & M = 12 \end{array}}$$

We have chosen $M = 12$ because this is the maximum number of any item that could be manufactured simply based on the materials constraints.

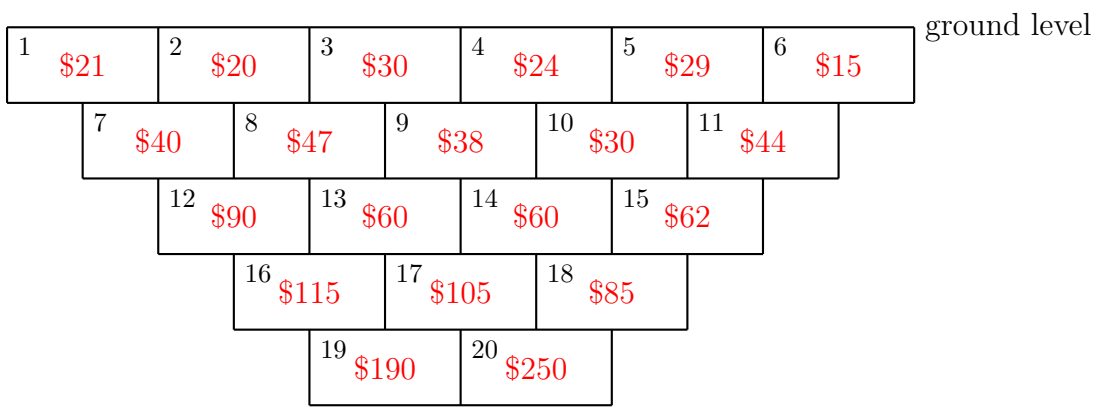
4.18 Pit Mine

This example illustrates simple dependency constraints in which an action cannot occur unless certain other actions also occur. Generalizations to sequential dependencies are treated in the exercises. This example also emphasizes the fact that sometimes objective or constraint coefficients are found only though computation from given data.

A prospective pit mine has been subdivided into twenty 25-ton blocks of in the following formation. The cost of removing each block is shown for each.



The only exception is block #1 which borders a river and would require the construction of a dike at a fixed cost of $3000. The extraction of any block cannot occur unless the two blocks directly above it are also extracted. The market value of extracted silver in $/ton of each block are as follows:



Formulate and solve an integer program to determine which blocks should be extracted to maximize profit. Assume that each block is either extracted in whole or not extracted at all.

Solution. Our decisions are whether or not to extract each individual block. Let

$$x_k = \begin{cases} 1 & \text{if block } k \text{ is to be extracted} \\ 0 & \text{if not} \end{cases}$$

for $k = 1, 2, \ldots, 20$. Let e_k be the extraction cost for block k, and let v_k be the market value per ton of material removed from block k. Then the objective function is

$$z = \sum_{k=1}^{20} (25v_k - e_k)\, x_k.$$

Notice that many of the higher level block extractions will result in negative profit, while many lower level block extractions are quite profitable. For example,

$$\begin{aligned} 25v_1 - e_1 &= 25(21) - 800 - 3000 = -3275 \\ 25v_2 - e_2 &= 25(20) - 800 = -300 \\ 25v_5 - e_5 &= 25(29) - 800 = -75 \\ 25v_{12} - e_{12} &= 25(90) - 1600 = +650 \\ 25v_{19} - e_{19} &= 25(190) - 2900 = +1850 \end{aligned}$$

The only constraints are the sub-excavation constraints which require, for example, that

$$\text{If } x_{10} = 1 \quad \text{then} \quad x_4 + x_5 = 2.$$

Because all decision variables are binary, we can enforce this condition as

$$x_4 + x_5 \geq 2x_{10}.$$

We see that if we wish to excavate block 10 ($x_{10} = 1$) then we must first excavate blocks 4 and 5 ($x_4 = x_5 = 1$). On the other hand, if $x_{10} = 0$ then x_4 and x_5 are free to be either 0 or 1. There are 14 such constraints, one for each of blocks 7 through 20. The integer program is

$$\boxed{\begin{array}{ll} \max\limits_{x} & z = c^T x \\ \text{s.t.} & Ax \geq 0 \\ & 0 \leq x \leq 1 \\ & x \in \mathbb{Z}^{20} \end{array}}$$

where

$$\begin{aligned} x &= \begin{bmatrix} x_1 & x_2 & \ldots & x_k & \ldots & x_{19} & x_{20} \end{bmatrix}^T \\ c &= \begin{bmatrix} -3275 & -300 & \ldots & 25v_k - e_k & \ldots & +1850 & +3350 \end{bmatrix}^T \end{aligned}$$

and

$$A = \begin{bmatrix}
1 & 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & -2 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & -2 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & -2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & -2
\end{bmatrix}.$$

The optimal solution is to extract all blocks except numbers 1, 7, 12, 16 and 19. The profit for such a strategy is \$1975. The cost of building a dike is not made up by high-profit excavations lower down. However, if the dike cost can be reduced to (for example) \$2750, then the optimal solution is to excavate all blocks with a corresponding profit of \$2025.

4.19 Team Building Strategy

This example demonstrates multiple simultaneous subset selection, or optimal division of a given set of objects, under a variety of constraints. Many illustrative examples of binary constraints are provided.

Big Bear Investments is creating teams of employees to create new profitable investment strategies. There are 18 employees, each of which has on file results of aptitude and teamwork ability tests. On a scale of 0 (low) to 100 (high), employee scores are shown in the table below. An employee who has been with the company at least 10 years is considered a senior employee and is indicated using bold face type.

	Amiel	Biniyah	Camua	***Dalliel***	***Eleth***	***Foculs***
Aptitude	59	93	69	81	86	50
Teamwork	67	32	56	35	31	66

	Gagavr	Hielaph	***Ikhael***	***Jhamue***	Kielach	Leraie
Aptitude	55	88	81	73	91	77
Teamwork	83	98	50	36	52	61

	Maliel	***Nuriel***	***Obor***	Peniyah	Quima	***Reilah***
Aptitude	30	99	37	71	91	65
Teamwork	64	50	59	93	52	38

Big Bear needs to form three teams satisfying the following conditions.

- Each team has three, four or five members.
- Each team must have a total aptitude score of at least 240.
- Each team must have an average teamwork score of at least 65.
- Senior employees cannot make up more than half of any team.
- No employee can be on more than one team.
- Neither Eleth nor Quima can be on a team that also includes either Biniyah or Dalleil.
- No subset of two or more from the list (Amiel, Eleth, Foculs, Jhamue) can work on the same team.
- If Leraie is on a team then Dalleil must be on the same team.
- If Camua and Kielach are on the same team then Heilaph cannot be on that team.

- Gagavr must be on one team.
- Either Eleth or Quima must not be on any team. (One or the other must be available to work elsewhere.)
- If Peniyah is on some team then Dalliel must also be on a team, not necessarily the same team.

Suppose individual productivity is defined as the product of aptitude and teamwork scores, and total productivity is the sum of individual productivity values. How can Big Bear Investments form exactly three teams that maximize total productivity?

Solution. We need to choose who will be on each of three teams. Let

$$x_{ij} = \begin{cases} 1 & \text{if employee } i \text{ is placed on team } j \\ 0 & \text{if not} \end{cases}$$

where teams are indexed by $j = 1, 2, 3$ and employees indexed by $i = 1, 2, \ldots 18$ in alphabetical order. Suppose the aptitude scores are a_i and the teamwork scores are t_i. Then we have the objective function

$$z = \sum_{i=1}^{18} \sum_{j=1}^{3} a_i t_i x_{ij}.$$

Consider the given constraints in the order presented. In each case, appropriate constraints are given without explanation. You should verify that the constraints enforce the given condition.

- Each team has three, four or five members.

$$3 \leq \sum_{i=1}^{18} x_{ij} \leq 5, \qquad j = 1, 2, 3$$

- Each team must have a total aptitude score of at least 240.

$$\sum_{i=1}^{18} a_i x_{ij} \geq 240 \qquad j = 1, 2, 3$$

- Each team must have an average teamwork score of at least 65.

$$\sum_{i=1}^{18} (t_i - 65) x_{ij} \geq 0 \qquad j = 1, 2, 3$$

- Senior employees cannot make up more than half of any team.

$$\sum_{i=1}^{18} x_{ij} - 2\sum_{i \in S} x_{ij} \geq 0 \qquad j = 1, 2, 3$$

where $S = \{4, 5, 6, 9, 10, 14, 15, 18\}$ is the index list for senior employees.

- No employee can be on more than one team.

$$\sum_{j=1}^{3} x_{ij} \leq 1 \qquad i = 1, 2, \ldots, 18$$

- Neither Eleth nor Quima can be on a team that also includes either Biniyah or Dalleil.

$$\begin{gathered} 2x_{2j} + x_{5j} + x_{17,j} \leq 2 \\ 2x_{4j} + x_{5j} + x_{17,j} \leq 2 \\ \text{for } j = 1, 2, 3. \end{gathered}$$

- No subset of two or more from the list (Amiel, Eleth, Foculs, Jhamue) can work on the same team.

$$x_{1j} + x_{5j} + x_{6j} + x_{10,j} \leq 1 \qquad j = 1, 2, 3$$

- If Leraie is on a team then Dalleil must be on the same team.

$$x_{4j} \geq x_{12,j} \qquad j = 1, 2, 3$$

- If Camua and Kielach are on the same team then Heilaph cannot be on that team.

$$x_{3j} + x_{8,j} + x_{11,j} \leq 2 \qquad j = 1, 2, 3$$

- Gagavr must be on one team.

$$x_{71} + x_{72} + x_{73} = 1$$

- Either Eleth or Quima must not be on any team. (One or the other must be available to work elsewhere.)

$$x_{51} + x_{52} + x_{53} + x_{17,1} + x_{17,2} + x_{17,3} \leq 1$$

- If Peniyah is on some team then Dalliel must also be on a team, not necessarily the same team.

$$x_{41} + x_{42} + x_{43} \geq x_{16,1} + x_{16,2} + x_{16,3}$$

The binary integer program has 54 decision variables and 48 constraints. An optimal team member selection is:

Team (#)	Aptitude (sum)	Teamwork (mean)	Team Members
1	261	65	***Nuriel***, Peniyah, Quima
2	272	65	Amiel, Gagavr, Kielach, Maliel, ***Obor***
3	296	65	***Dalliel***, ***Foculs***, Heilaph, Leraie

Notice that each of the three teams has a mean teamwork score of 65 (the minimum required) and total aptitude scores of at least 261 (above the 240 minimum).

4.20 DEA Solutions

When assessing the merit of a business or personal accomplishment, it is natural to compare against other similar entities. In this example, we introduce one such method, the technique of Data Envelopment Analysis.

DEA Solutions is a company that specializes in determining the best object from a given set of objects. Each object is associated with several data values, some of which are considered positive in nature and others negative. In a general sense, highly rated objects have strong positive scores and now negative scores. However, determining which object should be most highly rated may not have a clear answer because different objects can have different strengths and weaknesses.

DEA Solutions has been tasked with determining which player on the Spokane Indians baseball team in 2022 was the "best" position player. Each player is to be assessed based on their record in seven categories. The data set is given in the figure below. The first four categories measure player productivity (PA: plate appearances, TB: total bases, OPS: on base plus slugging percentage, R: runs scored) and the last three categories measure ways in which production is hindered (SO: strikeouts, CS: times caught stealing, DP: double plays grounded into). Only players who had at least 200 plate appearances are included in this analysis.

Player	PA	TB	OPS	R	SO	CS	DP
Julio Carreras	450	190	0.825	59	106	5	7
Bladimir Restituyo	423	161	0.708	55	87	10	7
Colin Simpson	373	175	0.900	53	78	1	3
Eddy Diaz	431	139	0.701	57	87	14	6
Drew Romo	420	139	0.693	52	81	3	8
Zac Veen	400	150	0.806	72	90	4	5
Grant Lavigne	282	113	0.875	41	68	3	3
Ronaiker Palma	247	91	0.737	28	27	3	5
Hunter Goodman	211	116	0.940	39	61	0	1
Robby Martin	302	96	0.611	30	92	4	6
Braiden Ward	225	50	0.616	43	58	5	0

DEA Solutions specializes in Data Envelopment Analysis in which each player is compared to all other players by maximizing a weighted productivity ratio. That is, for each player, k, maximize over all weights, w,

$$z_k = \frac{w_1(PA)_k + w_2(TB)_k + w_3(OPS)_k + w_4(R)_k}{w_5(SO)_k + w_6(CS)_k + w_7(DP)_k}.$$

However, the chosen weights must satisfy the constraints that for all other players this productivity ratio is no greater than 1. That is, for all $j \neq k$:

$$w_1(PA)_j + w_2(TB)_j + w_3(OPS)_j + w_4(R)_j \leq w_5(SO)_j + w_6(CS)_j + w_7(DP)_j.$$

Before we state the full optimization problem, we make three additional observations. First, it is good practice to "normalize" each data column to either the mean or maximum value of the column in order to help with numerical stability. Second, the objective is linearized by setting the denominator to equal 1. This is always possible because we can multiply the weights by any nonzero constant and achieve the same productivity ratio. And Third, we must restrict the weights to be non-negative so that positive aspects remain favorable and negative aspects remain detrimental to the productivity ratio. All together we have

$$\begin{aligned}
\max_{w} \quad & z_k = w_1(PA)_k + w_2(TB)_k + w_3(OPS)_k + w_4(R)_k \\
\text{s.t.} \quad & w_1(PA)_j + w_2(TB)_j + w_3(OPS)_j + w_4(R)_j \\
& \qquad \leq w_5(SO)_j + w_6(CS)_j + w_7(DP)_j, \;\; j = 1, 2, \ldots, 11 \\
& w_5(SO)_k + w_6(CS)_k + w_7(DP)_k = 1 \\
& w \geq 0 \\
& w \in \mathbb{R}^7
\end{aligned}$$

This LP will determine the productivity ratio of player k, with $0 \leq z_k \leq 1$. A similar problem must be solved for each player k where the inequality constraints are the same, but the equality constraint and objective are specific to the player of interest. These 11 linear programs yield the following results, where productivity scores are given as percentages ($p_k = 100 \times z_k^*$).

player	JC	BR	CS	ED	DR	ZV	GL	RP	HG	RM	BW
productivity (p)	76	80	100	85	89	100	86	96	100	59	100

Four players have maximum productivity percentages $p = 100$. These four players excel in different combinations of skills so that, compared to their peers, they add the greatest value to the team. Notice also that JC has a relatively low $p = 76$. This is likely due to the low run count and high strikeout total, or a combination of many factors.

4.21 Exercises

1. For the example of Alex's and Ryen's English paper, describe why the decision variable definitions "x_1 = Alex", "x_1 = Alex writes", "x_1 = time for Alex" are not acceptable.

2. Solve Alex's and Ryen's paper writing problem using the method of Vertex Enumeration. How many hours should each write? How many total grammatical errors will the paper have? What will be the effective length of the paper?

3. Model Alex's and Ryen's paper writing problem using the following list of decision variables. Is this formulation more or less difficult? Is the resulting LP more or less difficult to solve?

$$\begin{aligned}
x_1 &= \text{(number of hours that Alex writes)} \\
x_2 &= \text{(number of hours that Ryen writes)} \\
x_3 &= \text{(number of pages that Alex writes)} \\
x_4 &= \text{(number of pages that Ryen writes)} \\
x_5 &= \text{(number of errors that Alex makes)} \\
x_6 &= \text{(number of errors that Ryen makes)}
\end{aligned}$$

4. Consider the A-1 Furniture problem. Rework the problem to include the additional restriction that for tables and chairs to realize their profit potential there must be at least six times as many chairs manufactured as tables. Find the new optimal solution. Compare this solution with the previous formulation.

5. Consider the Greenside Yard & Lawn Care problem. Under the same grant and city restrictions, reformulate the problem in order to maximize the number of temporary jobs created under the restriction that no more than \$8000 is available to fund projects. Show that the maximum number of temporary jobs created is 29. What is the new optimal project strategy?

6. Reformulate the Tourist Trap problem to include the following additional variables

$$\begin{aligned}
x_5 &= \text{ total number of items purchased} \\
x_6 &= \text{ total number of T-shirts purchased} \\
x_7 &= \text{ total number of keychains purchased}
\end{aligned}$$

   Use these variables to simplify some of the existing constraints, at the expense of adding new constraints. Show that the solution is the same as the original formulation. Why might this modified formulation be preferable?

7. Solve the Woof&Meow pet food company problem. Suppose your job is to present your findings and recommendations to the company CEO. Compose one-page executive summary for your employer outlining the company's most profitable use of overstock resources. Include any findings which may be of interest to the company's decision makers.

8. Solve the Pack&Ship Delivery problem using the following data.

$$
\begin{aligned}
w &= \begin{bmatrix} 10 & 16 & 8 \end{bmatrix}^T \quad \text{(tons)} \\
s &= \begin{bmatrix} 6800 & 8700 & 5300 \end{bmatrix}^T \quad (m^3) \\
t &= \begin{bmatrix} 18 & 15 & 23 & 12 \end{bmatrix}^T \quad \text{(tons)} \\
v &= \begin{bmatrix} 480 & 650 & 580 & 390 \end{bmatrix}^T \quad (m^3/\text{ton}) \\
p &= \begin{bmatrix} 310 & 380 & 350 & 285 \end{bmatrix}^T \quad (\$/\text{ton})
\end{aligned}
$$

Determine which problem constraints are active constraints.

9. Consider the Shoes All Year production problem. Interpret the values of the optimal variables x^*, s^*, y^*, w^* in terms of the original problem statement.

10. Show that the stationary point for the least squares approach to the Thermo calibration problem is, in fact, a global minimizer of the objective function.

11. Based on the five-parameter model fit to the Mauna Loa CO_2 data, suggest a different model with no more than seven parameters that may more accurately represent the data set. Carefully justify your model. You do not need to solve this new problem.

12. The data provided in the table below represent measured data points (t_k, u_k) in a process $u(t)$ which is expected to be quadratic in time t. Use the method of least deviation to fit a general quadratic function to the data, then use this result to determine (a) the time t_{min} at which the minimum value of u occurs, (b) the minimum value of u, $u_{min} = u(t_{min})$ and (c) the value of u at time $t = 0$, $u_0 = u(0)$.

k	1	2	3	4	5	6	7	8	9	10	11
t_k	15	23	26	36	42	48	56	63	74	80	83
u_k	76	51	33	32	22	29	60	72	121	170	203

13. Suppose nodes d, e and t of the ExaByte problem represent computers in a single secure environment. If no data is allowed to leave this secure environment, what will the new data transfer model be? Solve this new model and compare the optimal solution to that of the original model.

14. Propose a different class of classifier functions that may perform well with the bank note data. Justify your choice.

15. Solve the Water Treatment Plant problem. Suppose your job is to present your findings and recommendations to your local state senator. Compose a one-page executive summary.

16. Modify the integer program formulation which tests single even integer instances of Goldbach's Conjecture to only allow sums of distinct primes. For example, $62 = 31 + 31$ is not allowed, but $62 = 3 + 59$ is allowed. Use your integer program to determine how many even numbers between 101 and 151 cannot be written as the sum of two distinct prime numbers.

17. Suggest and implement a modified Greedy Algorithm for solving the Book Packing problem. Your modified algorithm should try to make sure that all constraints are satisfied upon termination (if possible). Your new algorithm should never remove a book that is placed in the box.

18. Create a set of linear constraints that enforce the following condition on x:

$$\text{Either} \quad a \leq x \leq b \quad \text{or} \quad c \leq x \leq d.$$

    Build from the discussion in the CopyMaster example.

19. What is the (integer) minimum selling price of a glazed ceramic vase which makes it profitable to bring at least one on the trip? Keep all other problem parameters the same as in the original problem statement. Describe the new optimal solution for this vase selling price.

20. Describe (qualitatively) the solution to the fixed cost AJ Bread Distributors problem for the case $\min_{ij}\{e_{ij}\} \gg \max_{ij}\{c_{ij}\}$.

21. Consider the AJ Bread Distributors problem. Discuss the potential usefulness of creating a balanced problem through the use of a virtual bakery.

22. Solve the Puzzles and Games Inc. problem (a) ignoring the Either/Or manufacturing constraints and (b) including these constraints. Compare and discuss the results.

23. Justify the choice of binary decision variables in the Pit Mine problem. In particular, consider why it it not necessary to allow excavation of partial blocks.

Modeling Problems

24. A technician for a large animal shelter must formulate a diet from two commercial products, food A and food B, for the reptiles. In 200 grams of food A there are 16 grams of fat, 28 grams of carbohydrate, and 6 grams of protein. In 200 grams of food B there are 8 grams of fat, 50 grams of carbohydrate, and 10 grams of protein. The minimum daily requirements are 176 grams of fat, 980 grams of carbohydrate, and 200 grams of protein. Food A costs 8 cents per 100 grams and food B costs 18 cents per 100 grams. Write and solve an optimization model to answer the following question: What is the minimum cost diet that meets all daily requirements.

25. For a factory that manufactures trucks and cars, each vehicle must go through an assembly stage and a painting stage. If the painting was entirely devoted to cars, 500 per day could be painted. On the other hand, 400 trucks per day could be painted per day if the painting was devoted to trucks. The analogous numbers for assembly are 80 cars per day and 65 trucks per day. Each car produces $1300 profit and each truck produces $1500 profit. Write and solve an optimization model that determines how many cars and trucks the factory should produce to maximize profit?

26. An amateur veterinarian has determined that the minimum daily vitamin requirements for adult pigs is six units of vitamin C and three units of vitamin A. Barley sells for $1 per pound, and each pound contains one unit of each vitamin. Organic red turnips sell for $6 per pound, and each pound contains five units of vitamin C and 1 unit of vitamin A. Write and solve two different optimization problems that answer these two questions: (1) What is the least expensive recipe for pig food that satisfies the nutritional requirements? (2) What is the least expensive recipe that *exactly* satisfies the nutritional requirements?

27. Freya has been keenly observing her owner and would like to know the maximum number of kitty treats she can get in a day. There are a total of three different types of treats, chicken-flavored, beef-flavored, and catnip flavored. To get a chicken treats, she has found that it takes two meows. To get a beef treat takes 4 meows and to get a catnip treat takes 6 meows. Freya is sensitive to a healthy diet, so she must arrange her requests to have at least 10 but not more than 30 meat treats (chicken and/or beef). In addition, to she knows that she should not have more than one-third of the treats be catnip. But Freya will not be very happy if she does not get at least five catnip treats. Along with that, she is only content to meow up to 50 times per day – there must be some time for sleep. Construct and solve a mixed-integer program that determines a strategy for Freya to obtain the largest number of treats and also be happy and content.

28. You decide to travel to Portland to sell your remaining inventory of car parts. The weight, number in stock, and profit per item are listed in the table below. Airline regulations limit the total weight of your shipment to 214 pounds. Write and solve an optimization model to answer the following question: What is the maximum profit that can be realized on this venture and which parts should be taken to Portland?

item	weight(lb)	# in stock	profit($)
engine	104	2	151
transmission	62	4	68
seat	28	4	44
water pump	21	7	29

29. During each 4-hour period, the (24-hour) Red-Eye Diner requires the following number of on-duty staff:

```
00:00 - 04:00 : 3        12:00 - 16:00 : 5
04:00 - 08:00 : 8        16:00 - 20:00 : 9
08:00 - 12:00 : 6        20:00 - 24:00 : 8
```

Each person must work a shift of two consecutive 4-hour periods. Write and solve an optimization model that can be used to minimize the number of staff the the Red-Eye must hire. (Use only six decision variables.) Next, model and solve the modified problem in which no 4-hour period begins with a whole new crew coming on duty.

30. A supermarket can be staffed with full-time employees in any way that covers the labor requirements of 400 total person hours on Saturday, 400 on Sunday, 200 on Monday, 200 on Tuesday, 250 on Wednesday, 250 on Thursday, and 300 on Friday. Union rules require that each full-time employee work 5 consecutive full days (8 hours each) followed by 2 days off, but the supermarket can require them to work 1 day of overtime each week (with only 1 day off). A full-time employee costs the supermarket $10 per hour for the first 40 hours and $15 per hour on overtime days. Construct and solve optimization model that determines how the supermarket can meet its daily labor requirements at least cost.

31. A refinery has 5200 barrels of high quality oil (quality level 10) and 9800 barrels of medium quality oil (quality level 6). The refinery can make and sell two grades of gasoline from the raw oil: premium, which must have an average quality level of at least 8, and regular, which must have an average quality level of at least 7. There is no waste – the quantity of oil inputs equals the quantity of output

gasoline for both processes. Each dollar spent advertising premium results in 6 barrels of demand and each dollar spent advertising regular results in 15 barrels of demand, and there is no demand without advertising. Write and solve an optimization model to answer the following question: If premium is sold for \$70 per barrel and regular is sold for \$62 per barrel, what should the refinery manufacture in order to maximize profit?

32. A cattle feed producer makes two types of products using as ingredients barley and grass. Feed 1 must contain at least 75% barley and Feed 2 must contain at least 60% grass, both percentages by weight. Up to 1000 pounds of barley can be purchased at \$0.50 per pound, and up to 800 pounds of grass can be purchased at \$0.40 per pound. A customer is willing to purchase as much of both feeds that the producer can make, paying \$1.50 per pound of Feed 2 and \$1.35 per pound of Feed 2. However, the buyer requires a \$0.50 per pound discount on every pound of Feed 1 over 300 pounds. Formulate and solve a mixed-integer program that provides the producer a strategy for maximizing revenue.

33. Pots-R-Us obtains its stock of outdoor decorative pots from three suppliers. Each supplier sells pots in four sizes (large, medium, small and tiny) in the percentage quantities and prices given in the following table. Pots are purchased from each supplier in batches of 20. So, for example, each batch of 20 purchased from supplier 1 contains 7 large, 8 medium, 3 small and 2 tiny pots.

Supplier	Cost (\$/pot)	Large (%)	Medium (%)	Small (%)	Tiny (%)
1	4	35	40	15	10
2	3	30	25	35	10
3	2	10	10	50	30

Each quarter, Pots-R-Us places and order with each supplier. At least 100 large, 60 medium, 50 small and 40 tiny pots must be purchased each quarter in order to meet demand. Because the supplier facilities have limited production, at most 140 pots can be purchased from any supplier in any quarter. Write and solve an optimization model to answer the following question: How can Pots-R-Us minimize the cost of acquiring the needed pots for one quarter?

34. Villalovoz Inc. must allocate its 5 software engineers to a project. Each engineer has a different hourly rate and skill set. The project requires a total of 200 hours to complete. The total hours should include 60, 70 and 55 hours utilizing assigned engineers having skill sets 1,2,3, respectively. Write and solve a linear

program to minimize the total cost of completing the project given that company policy requires that each engineer work at least 30 hours. The hourly rates and skill sets for each engineer are given in the table below.

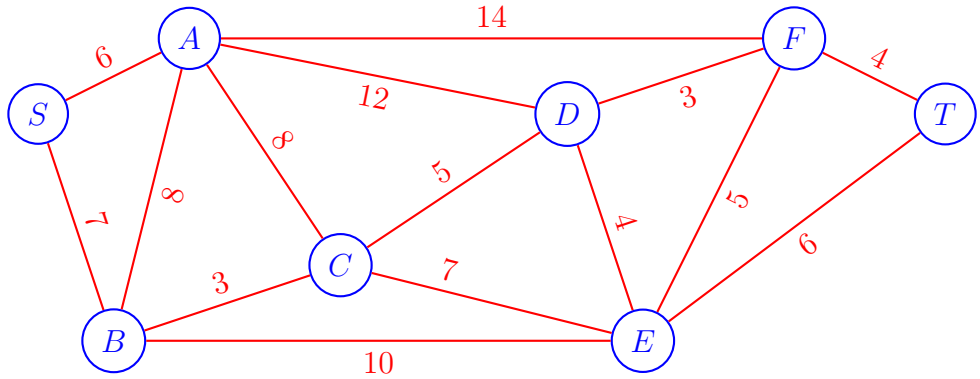
engineer	Hourly Rate ($)	Skill Set 1	Skill Set 2	Skill Set 3
1	32	yes	yes	no
2	27	yes	no	yes
3	35	no	yes	yes
4	39	yes	no	no
5	26	yes	no	yes

35. An investment manager has been tasked with providing a strategy for investing $1,000,000 in up to six mutual funds. These funds, the price per share, the expected return and the risk category are shown in the table.

fund	1	2	3	4	5
price per share ($)	100	41	75	20	25
expected return (%)	18	15	10	9	5
risk category	high	high	med	med	low

In order to control risk, management has specified that the dollar amount invested in medium risk funds must be at least 20% of the total investment and the dollar amount invested in low risk funds must be at least 10% of the total investment. Management has also stipulated that the investment must meet the diversification criterion that shares must be purchased in at least four of the funds, each with at least 10% of the total share purchase. With these guidelines, construct and solve and optimization problem that maximizes the expected investment return. *Be careful! "total dollar amount invested" is not necessarily the entire $1,000,000 dollars.*

36. Consider a network of roads that connects various common points as shown in the figure below. Here the circled letters represent, say towns, and lines represent roads connecting different towns. The numerical values show the drive time along each road between towns. Formulate and solve an integer program that determines the path from town S to town T of shortest total driving time. You may assume that the drive time within each town is negligible.



37. Revisit the road network problem (Exercise 36). Suppose that the added drive time within each town is a fixed time of 3. Formulate and solve this modified integer program.

38. A wood carver expects the following orders for tables: 5 in week one, 4 in week two, 4 in week three, 3 in week four. The wood carver already has 6 table tops in stock. During each week, the carver can produce 2 table tops at \$190 each, and can use overtime employees to make additional table tops if necessary, but at a cost of \$270 each. At the end of each week, a holding cost of \$25 per table top is incurred for any one held over for sale in the next week. Construct and solve an optimization model that minimizes the carver's cost for producing table tops that meet expected demands.

39. In an effort to reduce costs and simplify cooking, Tendai is considering a diet consisting of tofu, rice and broccoli. Some nutritional information and costs are shown in the table below. Compose an optimization model that can be used to provide a minimum cost daily diet that meets the following criteria: (a) at least 1600 calories, (b) at least 60 grams of protein, (c) no more than 40 grams of fat, (d) at least 100% DR (daily requirement) of iron and vitamin C. Then, solve your model and discuss the solution. Fractional serving amounts are allowed.

serving (1 cup)	cost (\$)	energy (cal)	protein (g)	fat (g)	iron (%)	vitamin C (%)
tofu	1.25	180	20	12	72	0
rice	0.10	200	4.3	0.4	1	0
broccoli	0.70	30	2.5	0.4	5	70

40. Comfort Deliveries ships sofas from three warehouses to four furniture outlet stores. Each store must receive 25 sofas this week. The shipping costs, in dollars per sofa, are:

		Store			
		1	2	3	4
	1	12	15	10	25
Warehouse	2	10	19	11	30
	3	21	30	18	40

Warehouses 1,2 and 3 have 20, 30 and 65 sofas available, respectively. For any sofa not shipped, there is a storage charge of \$7 per sofa at warehouses 1 and 2, and \$11 per sofa at warehouse 3. Labor contracts require that store2 must receive at least as many sofas from warehouse 2 as from warehouse 1. Management is participating in a quality control study which requires that store 4 receive at least half of its sofas from warehouse 3. Because warehouse 1 has only one delivery driver, its stock can be delivered to at most one store. Formulate and solve an integer program that provides a minimum cost shipping strategy for Comfort Deliveries that satisfies all given constraints.

41. Five assembly workers are to be assigned to five different stations, the table below gives the productivity (on a scale of 1-10) of each worker at each station.

Worker	parts check	fitting	reinforcing	painting	finishing
1	1	6	6	5	9
2	3	6	9	4	10
3	2	5	6	3	6
4	3	6	6	2	7
5	4	7	7	2	8

Formulate and solve an optimization model to answer the following question: If the total productivity of the assembly line is the sum of the productivity values of the workers assigned to the stations, what assignment of workers to stations maximizes total productivity?

42. Foreign dignitaries from r countries are visiting your humanitarian aid headquarters in the United States in order to decide whether or not to financially assist your operations. You have t language translators to assign as hosts to your visitors. Each translator has completed a test of proficiency in the language and customs of each visitor. The score of translator i in the language of visitor j is s_{ij}, where $i = 1, 2, \ldots, t$ and $j = 1, 2, \ldots r$. Formulate a linear or integer program that can be used to assign a translator to each visitor based on maximizing the sum total proficiency scores of the assigned pairs. Assume

$t \geq r$. Exactly one translator must be assigned to each visitor. No translator can be assigned to more than one visitor.

43. Revisit the foreign dignitaries problem (Exercise 42). Construct a linear or integer program that adheres to the same constraints but maximizes the minimum proficiency score among all assigned pairs.

44. A car rental company would like to measure the efficiency of five of its rental centers: Seattle, Vancouver, Spokane, Pasco, Walla Walla. This is done via "data envelopment analysis," which compares ratios of weighted sums of outputs for a center to weighted sums of inputs for the same center. The efficiency of any distribution center is defined to be the maximum ratio over all choices of (non-negative) weights for which none of the corresponding five rations (one for each center) exceeds 1. The table below shows the average daily inputs and outputs of each distribution center. Write and solve five optimization models to find the efficiency of each of the five distribution centers.

Outputs	Seattle	Vancouver	Spokane	Pasco	Walla Walla
rentals (car-hours)	1200	1100	1300	1500	1400
advance bookings (#)	20	16	21	38	40
Inputs	**Seattle**	**Vancouver**	**Spokane**	**Pasco**	**Walla Walla**
Complaints (#)	11	10	11	12	12
Maintenance Cost ($)	2000	1350	3000	4200	4500

45. Marble Construction is remodeling an office building. There are 12 tasks which cannot be completed simultaneously as some tasks can not be started until other tasks have been completed. The table shows each task, the time required to complete the task, and the tasks which must be completed before starting the task. Suppose that tasks can be worked concurrently as long as tasks dependencies are met. Formulate a linear or integer program that can be used to find the minimum duration of the remodel and the earliest time each task can be started.

task #	task	time (days)	prior tasks
1	Remove old furniture	12	(none)
2	Wall remodel	8	1
3	New duct-work	14	2,4
4	New elevators	21	1
5	Upgrade fire escapes	6	(none)
6	Wiring	15	3
7	Lighting	9	2,3,6
8	Windows	6	2,5
9	Landscaping	12	5,8,11,12
10	Entryway	7	12
11	Exterior painting	8	5,8
12	Install new furniture	23	4,6,7

46. Radio station KOPT is studying where to place radio towers in the hills around Dennis County. Four tower sites are under consideration. The costs of building on sites A, B, C and D are \$3500, \$4000, \$4500 and \$2700, respectively. A tower on each site can reach one or more of ten districts of varying populations as shown in the table below, where populations are shown in thousands. For example, sites B and D can both broadcast to district 7 which has a population of 12,000. Construct and solve an integer program the provides the maximum radio coverage, by population, if KOPT has \$10,000 for construction. Be careful to count any covered district only once.

district:	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
site A:	30	12		21	13	4				
site B:		12	28		13	4	12	20		
site C:				21	13	4		20	11	25
site D:						4	12	20	11	

47. You are considering which of ten acquaintances to invite to a party: Abdul, Baji, Carlos, Daniel, Esther, Fouzia, Goro, Hemene, Ingrid and Jie. Write and solve an optimization model that maximizes the number of guests under the following conditions.

    (a) At least two women (Baji, Esther, Fouzia, Ingrid, Jie) and at least two men (Abdul, Carlos, Daniel, Goro, Hemene) should be invited.

(b) Fouzia and Daniel should not both be invited.

(c) If Abdul is invited then Jie should also be invited.

(d) If Baji and Daniel are invited then Hemene should also be invited.

(e) If Esther and Jie are both invited then neither Goro nor Abdul should be invited.

48. You are trying to decide which of seven possible classes to take next semester. Your choices (i), their credit hour load (c_i), and the grade points per credit hour you think you can achieve (g_i) are given in the table below. Write and solve an optimization model that can be used to maximize the number of credits you can earn while keeping your grade point average (GPA) not below 3.5. Furthermore, if you take class 3 you must also take class 5.

class (i)	credit hours (c_i)	grade points per credit hour (g_i)
1	3	4.0
2	3	3.3
3	4	4.0
4	4	3.0
5	3	2.7
6	3	3.7
7	4	3.0

49. Consider a monetary system with only 1¢, 5¢, 10¢, 20¢, 25¢, 50¢ and \$1 coins. A sales clerk would like to know how to provide change totaling a given amount d¢using the fewest number of coins. However, because 25¢ and \$1 coin are very similar in size and shape, combinations of these coins should be avoided. For example, \$1.30 cannot be formed using a \$1 coin, a 25¢ coin and a 5¢ coin, but can be formed using a \$1 coin and three 10¢ coins. Formulate an IP that can be used to solve this problem. What is the minimum number of coins required in order to form 99¢ in change?

50. Apples-2-Apples Management would like to create two teams from a group of m employees. Productivity scores for each employee are known values p_k, $k = 1, 2, \ldots, m$. Formulate an integer program which A2A can use to divide these employees into two teams so that the total productivity scores are as similar as possible.

51. A chemical company manufactures two products from four raw materials. The company can use three different batch processes in any combination to meet consumer demand. The table below shows, for each process, the required raw materials, the amount of output products and the manufacturing cost. The table also includes the consumer demand for each product. Formulate an integer program that can be used to find the processes to employ that minimizes cost to the company while meeting consumer demand. The available raw material quantities are 125, 175, 225 and 150, respectively.

	Process 1	Process 2	Process 3	Consumer Demand
Raw Material 1	1	2	0	
Raw Material 2	2	2	1	
Raw Material 3	4	1	2	
Raw Material 4	0	3	4	
Product 1	3	2	1	217
Product 2	1	3	1	167
Cost	\$44	\$32	\$29	

52. A producer of trail mix has year-end left-over stock of peanuts, chocolate, raisins, pretzels and dried papayas. They would like to make and sell a holiday mix from this stock. The goal is to create a mix of these ingredients that has as close to 240 calories per bag as possible. Processing machinery dispenses each ingredient in integer unit amounts. The following table describes the stock.

Item	Weight (grams per unit)	Calories (calories per unit)
peanuts	6	35
chocolate	5	27
raisins	7	21
pretzels	4	15
papaya	8	4

Any optimal recipe for trail mix must contain at least one unit of each of the five items. The total bag weight cannot exceed 57 grams (about 2 oz.). Write and solve an optimization model whose solution provides the optimal trail mix recipe.

53. A breakfast cereal manufacturer wishes to make a new product from some combination of the five ingredients: oats, wheat, flax, honey and blueberries. The company machinery dispenses these ingredients in unit quantities into a large mixing vat. Oats, wheat and berries are dispensed in units of cups; honey and flax are dispensed in units of tablespoons. The table shows the calories contained in one unit of each ingredient. Write and solve an optimization model to find the mixture of ingredients that most closely provides a mixture of 2500 calories under the additional constraints that (a) at least on unit of each ingredient must be used, (b) no more than 7 units of any ingredient can be used and (c) the total volume of ingredients cannot exceed 9 cups.

	oats	honey	wheat	flax	blueberries
unit	1 cup	1 tbsp	1 cup	1 tbsp	1 cup
calories	158	64	651	55	85

54. A widget manufacturer ships 12,400 widgets to customers every month. Widgets may be produced at seven different factories. The production capacity, cost per widget and fixed costs for factory operation are given in the table below.

Factory	Fixed Operating Cost ($)	Fixed Dormant Cost ($)	Widget Cost ($/widget)	Capacity (# widgets)
1	40,000	4,000	20	3000
2	20,000	3,300	25	2400
3	16,000	3,500	30	2800
4	31,000	2,900	28	2100
5	22,000	2,900	29	3100
6	34,000	4,600	24	2500
7	30,000	3,100	31	5000

The fixed operating cost for any factory is only incurred if that factory makes widgets, otherwise a fixed dormancy cost is incurred. Write and solve a linear optimization model whose solution provides the optimal strategy for minimizing total cost while meeting monthly customer demand.

55. A farmer is planning an orchard of mixed apple, pear and cherry trees that can hold a maximum of 350 trees. The seasonal cost of labor per tree is $150, $200 and $250, respectively. The seasonal cost of materials per tree is $275, $175 and $125, respectively. Planting logistics require that if any apple trees are planted

then at least 150 must be planted. Similar limits on pear and cherry trees are 50 and 80, respectively. Write and solve an optimization model that maximizes the number of trees which the farmer can plant when labor costs are limited to a maximum of \$50,000 and materials costs are limited to a maximum of \$60,000.

56. Pears-2-Pears Management would like to create two teams from a group of 20 employees. Productivity scores for each employee are known values p_k, $k = 1, 2, \ldots, 20$. Either both teams must have between 3 and 5 members or both teams must have between 7 and 9 members. The teams do not need to have the same number of members. Formulate an integer program which P2P can use to two teams whose total productivity scores are as similar as possible.

57. Determine the best fit quadratic polynomial fit for the Thermo data by two methods: least squares and least deviation, and discuss your results.

58. Consider the polyhedron $P = \{x \in \mathbb{R}^2 \mid Ax \leq b\}$.

    (a) Construct a linear program that can be used to find the largest horizontal line segment contained in P.

    (b) Construct a linear program that can be used to find the center $z \in \mathbb{R}^n$ and radius $r \in \mathbb{R}$ of the largest ball contained in P.

    (c) Construct a linear program that can be used to find the rectangle of largest perimeter contained in P. Assume that the sides of the rectangle are oriented parallel with the coordinate axes.